

QuPath: A great tool for spatial omics image and data analysis

Part I: Multiplexed immunofluorescence image analysis

SPAOM 2026 | Community Workshop 6 | Bioimage Analysis and Spatial Omics

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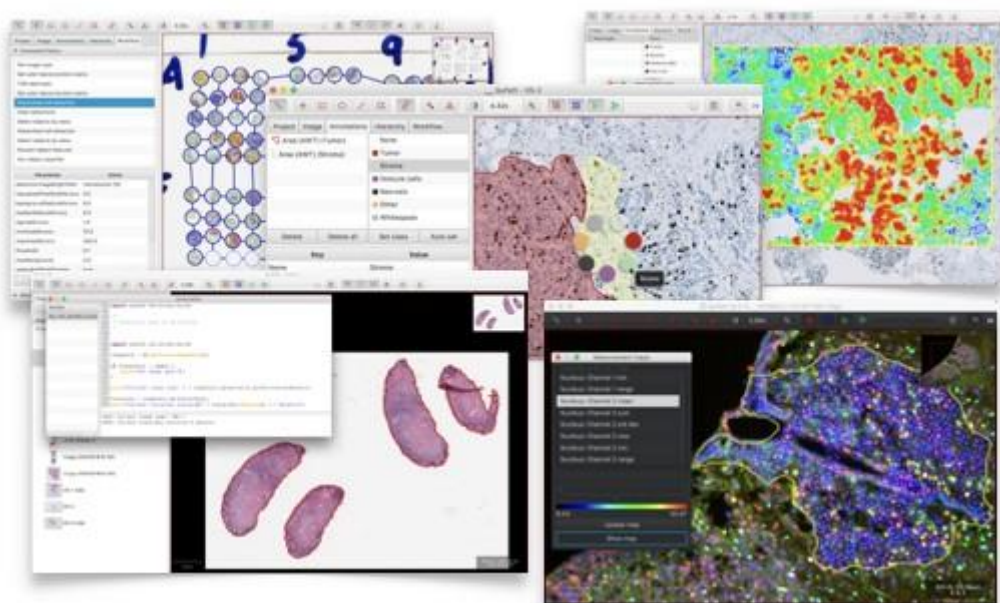


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Practical Exercise. Cell Phenotyping and Spatial Analysis of Multiplexed Images



DATA ▶ Multiplexed high-plex fluorescence image of a melanoma tissue sample containing 19 channels.

The panel includes a nuclear stain and markers associated with immune-cell lineages, tumor cells, stromal components, vascular structures, cell proliferation, cytotoxic activity, antigen presentation, and immune-checkpoint expression.

The multiplex panel comprises:

- **Nuclear identification and segmentation:** DAPI
- **Immune-cell identification:** CD45
- **T-cell phenotyping:** CD3 and CD8
- **B-cell identification:** CD20
- **NK-cell identification:** CD56
- **Myeloid and antigen-presenting cell characterization:** CD11c, CD68, CD14, and HLA-DR
- **Immune activation and cytotoxic function:** PD-1 and Granzyme B
- **Immune-checkpoint expression:** PD-L1
- **Tumor and melanocytic characterization:** CK and S100
- **Proliferation:** Ki67
- **Stromal and mesenchymal characterization:** αSMA and Vimentin
- **Vascular identification:** CD34

FILE ▶ The image is provided as a **Ome.tiff** file:

- **melanoma_cropped.ome.tiff:** cropped lightweight image, 2.0 GB, strongly recommended for the Community Workshop.

Important: The raw image is a full 66.9 GB image which requires a computer that meets the **minimum hardware** requirements. To ensure a smoother and more consistent workshop experience, all participants are strongly encouraged to use **melanoma_cropped.ome.tiff**.

Target Immune Populations ▶

Phenotype	Marker Combination	Biological Role
T Cells	CD45+ CD3+CD8-	Total T-lymphocyte population involved in adaptive immune responses.
CD8+ T Cells	CD45+ CD3+ CD8+	Cytotoxic T cells that recognize and eliminate tumor cells.

GOAL ▶ To perform cell-level phenotyping of immune populations in a multiplexed fluorescence image using **QuPath**. This exercise will focus on the identification of three hierarchically related T-cell populations:

- **T Cells:** CD45+ CD3+CD8-
- **CD8+ T Cells:** CD45+ CD3+ CD8+

Participants will explore two complementary phenotyping strategies:

1. **Threshold-based classification**, using QuPath **single measurement classifiers** to assign marker positivity according to cell-intensity thresholds.
2. **Supervised machine learning classification**, using manually labelled training examples and multiple cellular measurements to classify immune phenotypes.

The threshold-based approach provides a simple, transparent, fast, and reproducible method for identifying populations with clearly separated marker expression. The supervised machine learning approach offers greater flexibility, considers multiple features simultaneously, and may be more robust when dealing with heterogeneous, overlapping, or noisy cell populations.

Threshold-Based Classification	Supervised ML Classification
Simple and intuitive	More flexible
Transparent decision rules	Uses multiple features simultaneously
Fast to optimize	More robust against biological heterogeneity
Easy to reproduce	Better suited for noisy or complex populations
Ideal for well-separated markers	Can capture subtle phenotypic differences

The exercise will also include the identification of the **tumor compartment** using the S100 signal. This compartment will serve as a reference region for downstream spatial analyses, including:

- Quantification of immune-cell infiltration within and around the tumor.
- Measurement of immune-cell distances to the tumor border.
- Measurement of distances between the identified T-cell populations.
- Analysis of cellular neighborhoods and spatial proximity.
- Generation of cell-density maps and spatial heatmaps.

By the end of the exercise, participants will be able to compare rule-based and data-driven phenotyping strategies, generate spatial maps of immune populations, and examine the organization of the immune microenvironment in relation to the tumor compartment.

Workshop Materials and Data Access

All files required for this practical session will be distributed through the following GitHub repository. Replace the placeholder below with the final public repository address before distribution.

Repository: https://github.com/cnio-cmu-BioimageAnalysis/cw6_spaom_2026

The repository contains the files, scripts and supporting resources needed to reproduce the QuPath workflow presented during the workshop. The docs directory includes the following documents:

- SPAOM2026_QuPath_Practical_Guide.pdf, containing the detailed step-by-step instructions for the practical session.
- CW6_QuPath_Installation_Guide.pdf, containing the software installation and environment preparation instructions.
- QuPath_SPAOM2026_Community_Workshop_PartI_final.pdf, containing the introductory presentation for Part I of the workshop.

Recommended local structure: keep the image data, Cellpose script, pre-generated GeoJSON detections, and QuPath project in a single workshop folder. Do not move the source image after importing it into the project, because QuPath stores a link to the image location rather than copying the image into the project.

Before the workshop: download the repository contents and the 2.0 GB `melanoma_cropped.ome.tiff` image in advance. Confirm that QuPath opens the image and that the image type is set to Fluorescence.

Important: The GitHub repository should contain the practical guide, scripts, pre-generated segmentation file, classifier resources, and a README with software versions, installation instructions, data-download links, expected folder structure, and licensing or citation information.

1. Create a Project

To start our analysis, it is necessary to create a QuPath project. To do so, follow these steps:

- Create an **empty folder** with the name *cw6_qupath* to create your project location. All automatically generated files will be stored here, following QuPath architecture.

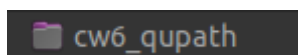


Figure 1. Empty folder creation for the QuPath project.

- Open the QuPath application
- Go to the Project tab

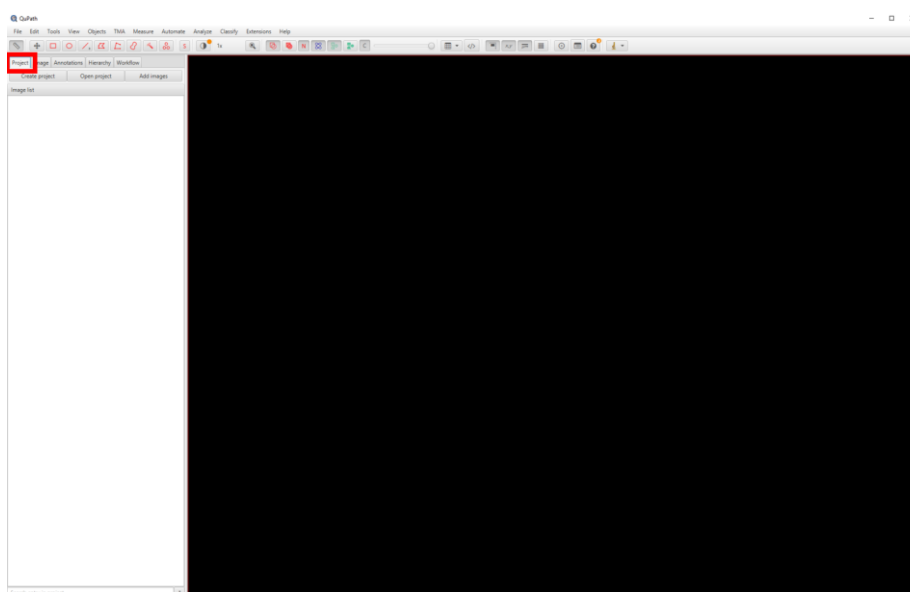


Figure 2. QuPath project tab (rounded in red)

- Click **Create a project**

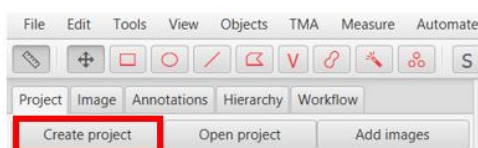


Figure 3. Create project button inside the Project tab.

- When the pop-up window appears, browse and choose the empty folder that you have just created and click **Select folder**.

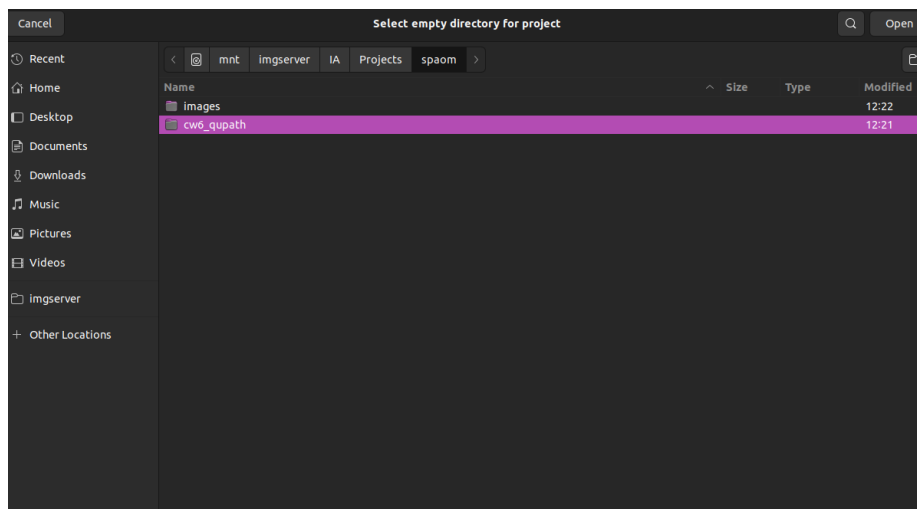


Figure 4. Selecting the empty folder to create the QuPath project.

- Congrats! Now you have your QuPath project created:

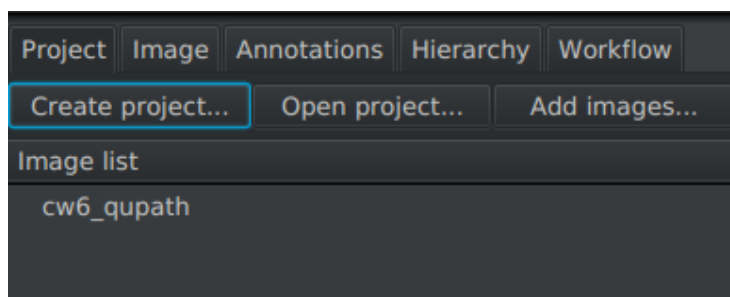


Figure 5. Created QuPath project.

Note: When creating a project, QuPath will automatically generate all files within your project folder. These files follow typical QuPath architecture, needed for the software to interpret your project. Among all of them, the project.qpproj file is the most important file so QuPath can reopen your project later.

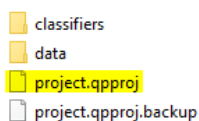


Figure 6. Typical QuPath project architecture.

1. Importing Images into a Project

Once the project is created, it is time to import the images we want to analyze. There are **two ways** to **add images** to a **QuPath project**:

Note: You could do it either way. We **recommend you do it in both to get knowledge** about the different options QuPath has, **but in practice you would only need to do it by one way**

A) Using **Add images**

- Go to **Project → Add images → Choose files** and browse the image file 'melanoma_cropped.ome.tif'



Figure 7. Importing images to the project using "Add images" button.

- Select the images and click **Open → Import**
- Double click on the image
- Select **Fluorescence** under **Set image type** in the prompt window and click **Apply**.

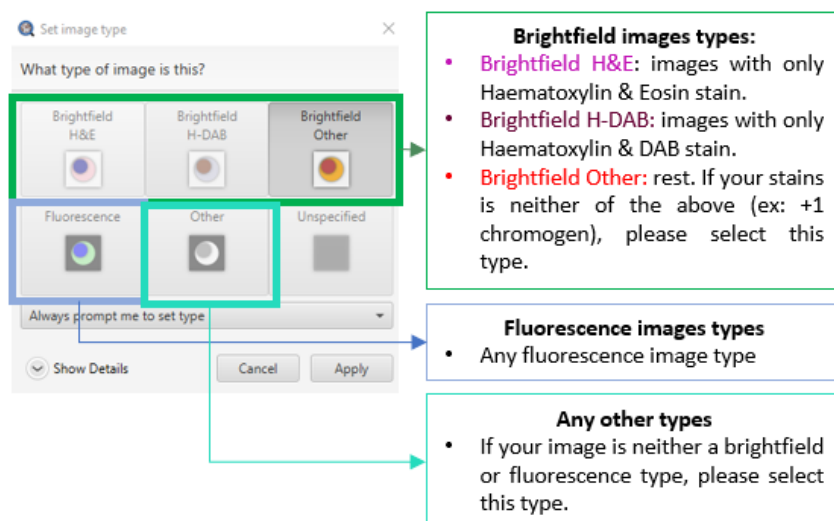


Figure 8. Image types in QuPath.

Note: It is **very important** to define the **image type** correctly as it will have an impact on further segmentation for analysis.

B) Drag and drop

- Simply select all the files you want to import (**melanoma_cropped.ome.tiff**) and drag image files into the QuPath workspace.
- Select *Fluorescence* under **Set image type** in the prompt window and click **Apply**.
- QuPath will prompt you to confirm adding them to the project.

Note: In this case it is still **very important** to define the image type correctly as mentioned before.

Tip (not necessary for this exercise): The **images need to always be at the same location**. QuPath does not save the images in the project folder, instead it links the project data to the image location. If you change the image location and try to open the project you would see a prompt asking you to enter the image new location (if images are not being found by the software, they will appear in red). Sometimes, QuPath will be able to find the new location, in which case, the new path will appear to the right of the Original URI; otherwise, you can simply click on **Search...** → browse the new location and click **Open** → **Check** that all errors are solved (images are found again) → click on **Apply changes**.

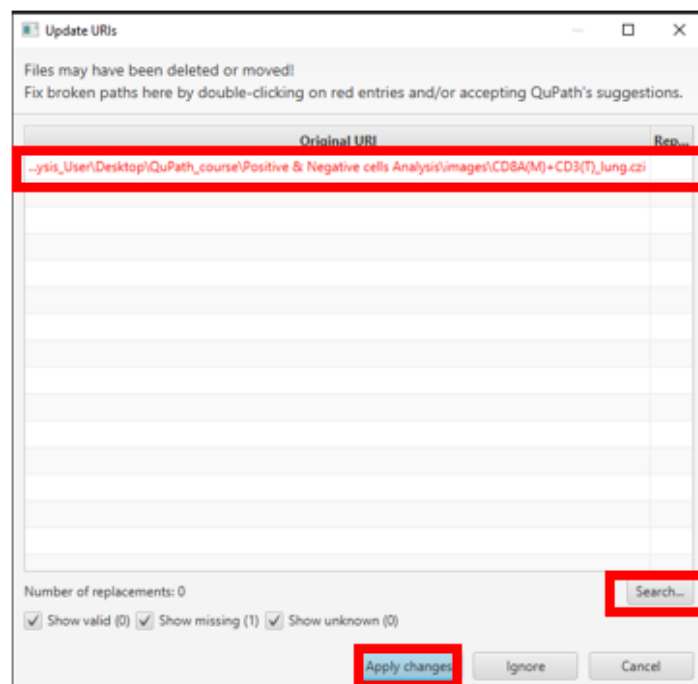


Figure 9. Updating image path after changed to a new location.

2. Nuclei & Cell Detection

Visual Inspection of Nuclear Signal

- Open **Brightness & Contrast**  and activate only the **DAPI (C1)** channel.
- This allows you to inspect nuclear morphology and estimate appropriate parameters for nucleus-based cell detection.

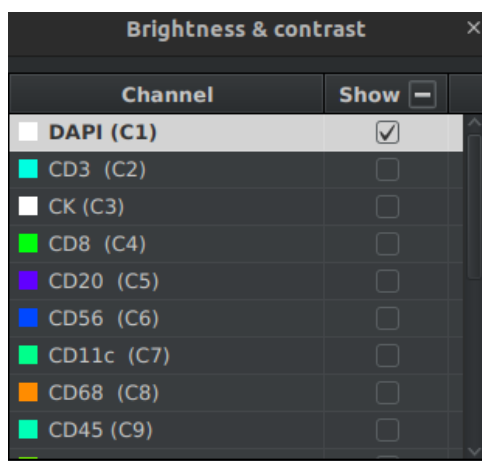


Figure 10. Brightness & Contrast functionality with DAPI channel activated.

Estimate Nuclear Diameter

- Open **melanoma_cropped.ome.tif** in the current QuPath project.
- Select the **Line annotation tool**  and zoom in to individual nuclei.
- Draw several line **annotations** corresponding to nuclei of different shapes and sizes.

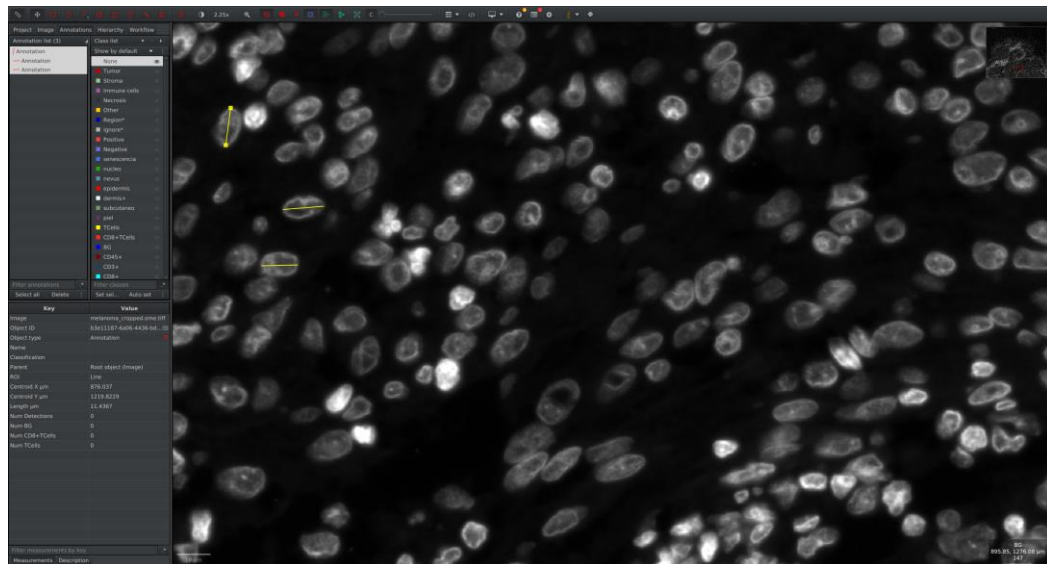
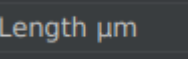


Figure 11. Different annotations on diverse nuclei.

- After drawing several **representative nuclei**:
 - Go to the **Annotations** tab.
 - Select each annotation one by one.
 - Inspect the **Length µm** measurement shown in the left-lower panel. 
- These values are useful to define the detection range for cell segmentation. Once you have reviewed the annotations, select all of them and delete them.

Cell Segmentation using Cellpose

Before performing cell phenotyping, individual cells will be segmented using **Cellpose extension in QuPath**. A preconfigured Cellpose script, including the model and segmentation parameters, will be provided in the workshop materials under: `/cw6_spaom_2026/cw6_qupath/scripts/cellpose_segmentation.groovy`

- Having the image **'melanoma_cropped.ome.tiff'** opened in QuPath.
- Press **Ctrl + L** and search for **Create full image annotation**.

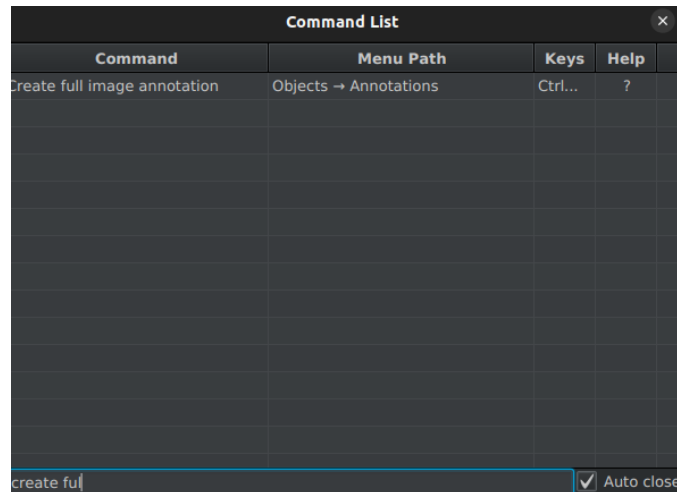


Figure 12. Command List filtered to the Create full image annotation command.

3. Run the command and confirm that an annotation covering the entire image has been created.
4. Press **Ctrl + L** again and search for **Script editor**.
5. Drag & drop the Cellpose script (**cellpose_segmentation.groovy**) for nuclei segmentation located at **/cw6_spaom_2026/cw6_qupath/scripts/cellpose_segmentation.groovy** into the Script Editor.

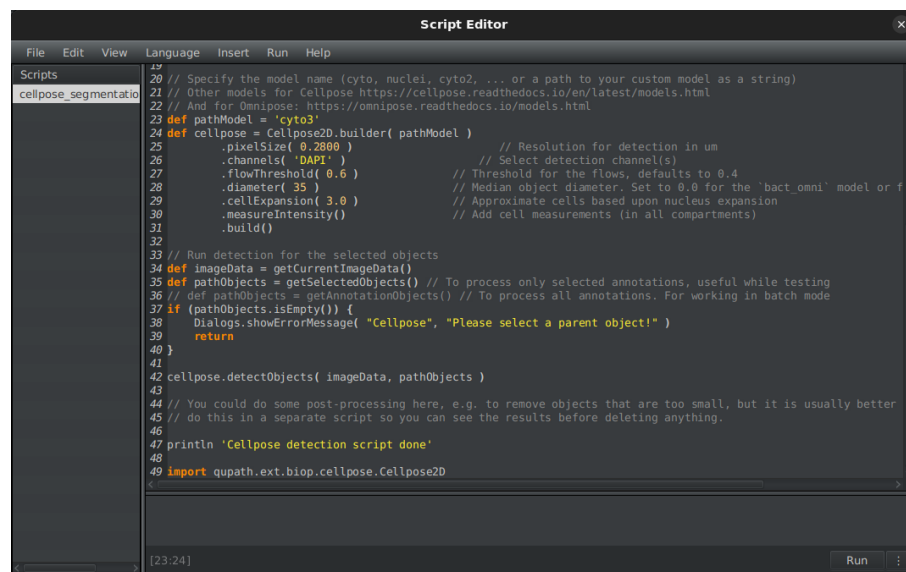
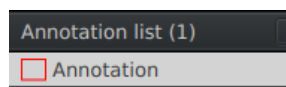


Figure 13. Open the provided Cellpose script (cellpose_segmentation.groovy) from the path.

Note: These parameters were selected after visually inspecting representative nuclei in the DAPI channel. The pixel size reflects the image calibration, the diameter corresponds to the approximate nuclear size, and the cell-expansion value provides an estimated cellular compartment for subsequent marker-intensity measurements and cell phenotyping.

6. Ensure that the **Full image annotation** is selected.



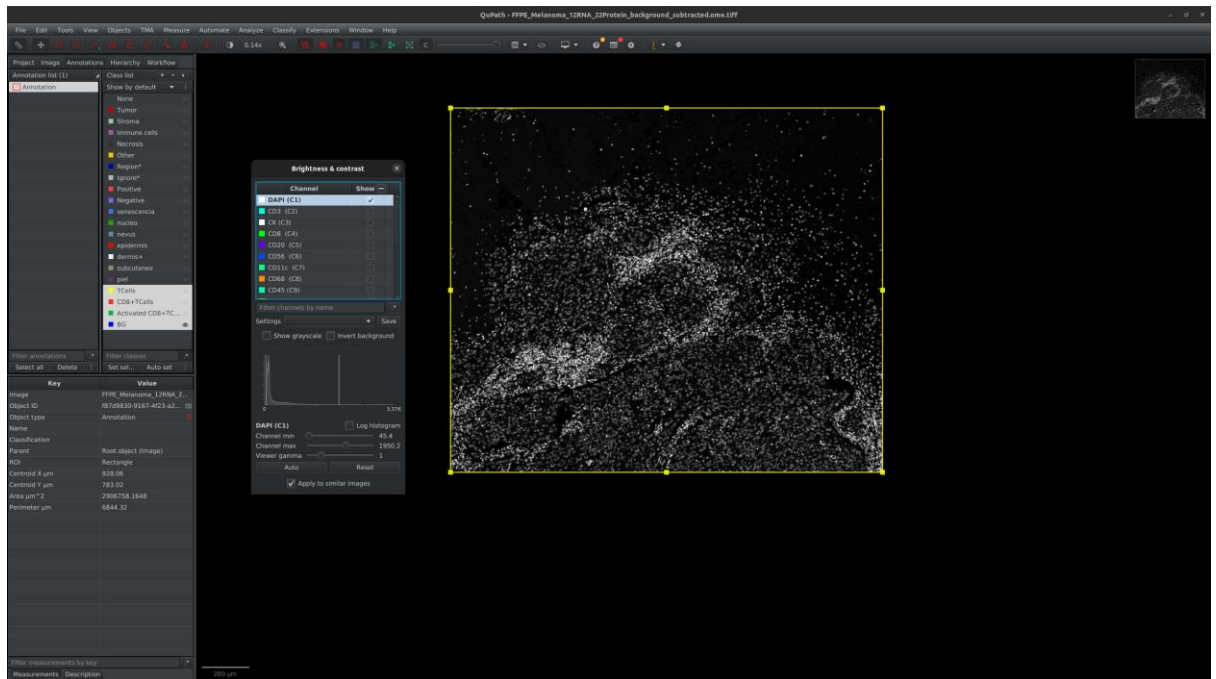


Figure 14. Full-image annotation selected as the parent region for Cellpose segmentation.

- Before running the script, verify that the command `measureIntensity()` is **uncommented**. This is essential because it calculates marker-intensity measurements for the segmented cells. These measurements are required for subsequent phenotyping using both **Single Measurement Classifiers** and **supervised ML classifiers**

- Run the script.

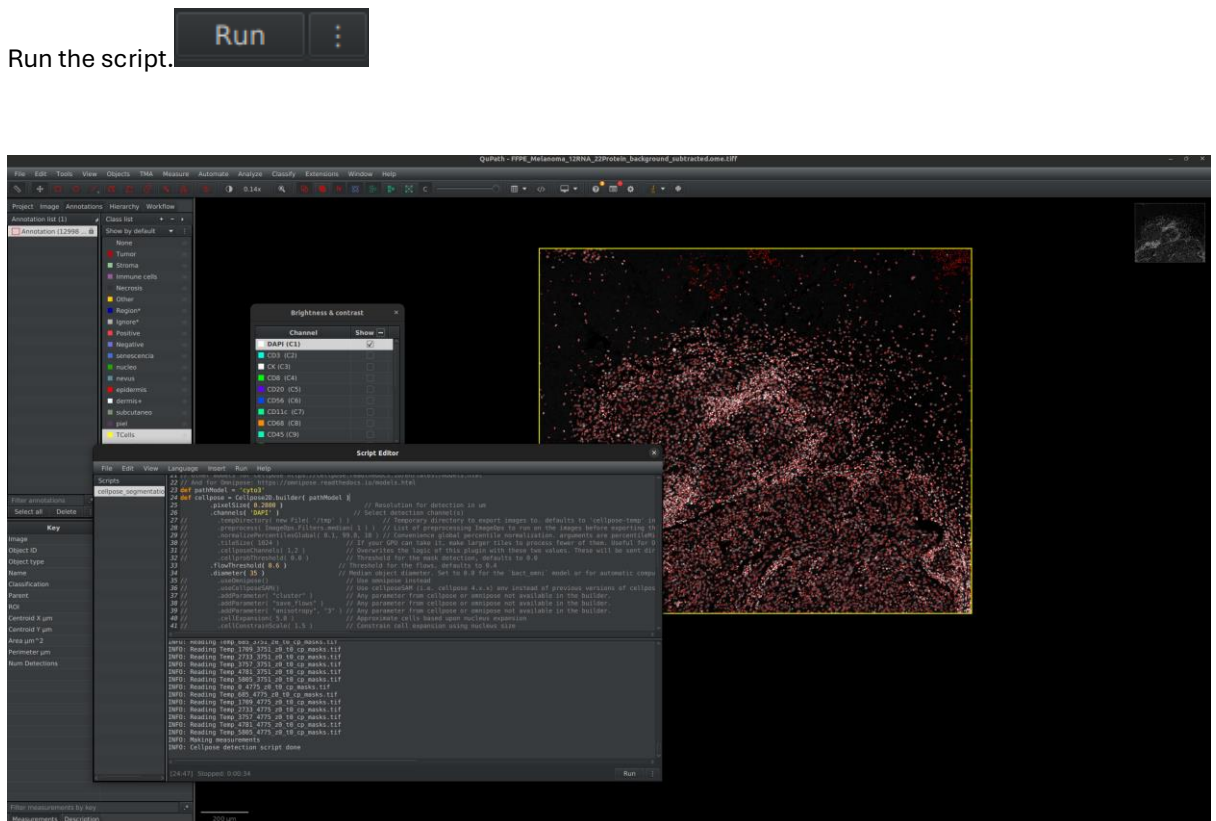


Figure 15. Cellpose script execution with intensity-measurement calculation enabled for downstream phenotyping.

Alternative workflow using pre-generated Cellpose segmentations

If Cellpose cannot be run on your computer, Cellpose segmentation masks will be provided as a **.geojson** file (**melanoma_cropped.geojson**) located at **geojson_masks** folder under **/cw6_spaom_2026 /cw6_qupath** project. These masks were generated in advance using the same image and Cellpose parameters described in this exercise.

Importing these masks into QuPath is straightforward: simply **drag and drop the .geojson file directly onto the image in the QuPath viewer**. The pre-generated segmentation objects will then be imported into the image, allowing participants to continue with the downstream steps of the workflow.

Important: Make sure that the **.geojson file** is imported onto the correct image, since the coordinates of the segmentation objects correspond specifically to that image '**melanoma_cropped.ome.tif**'

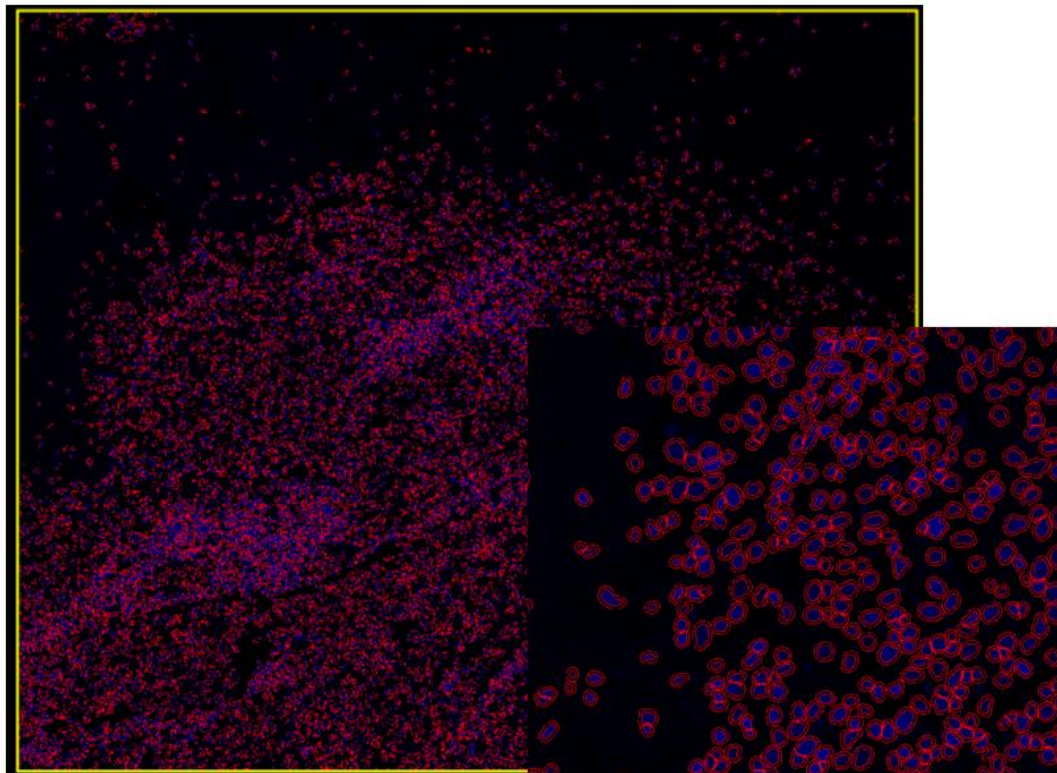


Figure 16. Cellpose-derived cell detections across the full workshop image, with a magnified view showing nuclear and expanded cellular boundaries.

3. Cell Phenotyping Using Single Measurement Classifiers (Threshold-based Workflow)

Create the Cell-Phenotypes Classes

- In the **Annotations** panel:
 - Click on the **+** button on the **Class list** menu to add a new class

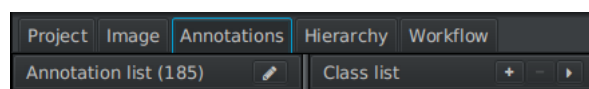


Figure 17. Class list controls used to add phenotype classes in QuPath.

- Create the following classes:
 - **CD45+**
 - **CD3+**
 - **CD8+**
 - **CD8-**

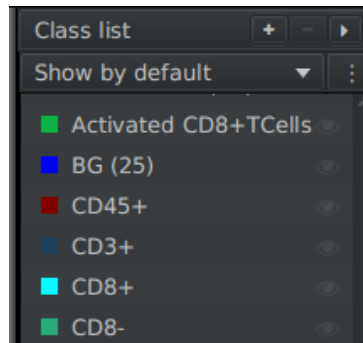


Figure 18. Edit class dialog used to assign a display color to an object class.

If these classes already exist, you can skip this step. You can edit the display colors by double-clicking on each class name.

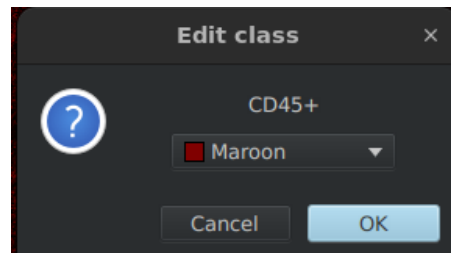


Figure 19. Editing the display color of an object class in the QuPath Class list.

Create CD45+ Classifier

- Activate only the **CD45** channel in **Brightness & Contrast**  and adjust the display range so that positive cells are clearly visible.

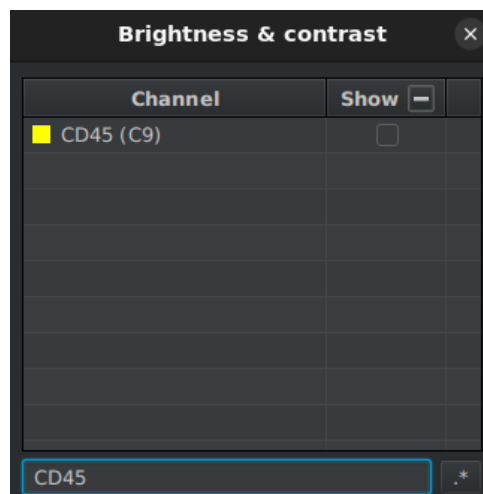


Figure 20. Brightness & Contrast control used to adjust channel visualization in QuPath.

Note: To quickly locate the CD45 channel, use the “Filter channels by name” search bar in the Brightness & Contrast window and enter CD45. This is particularly useful when working with highly multiplexed images containing many channels.

- Go to **Classify > Object classification > Create single measurement classifier**
- Configure the classifier using the values shown in the exercise figure for **CD45+** (Figure 87).

Note: Check **Live preview** to visualize how the selected threshold affects cell classification in real time and how it provides a proper separation between positive and negative cells.

- The click on **Save** to save the classifier as **CD45+**.

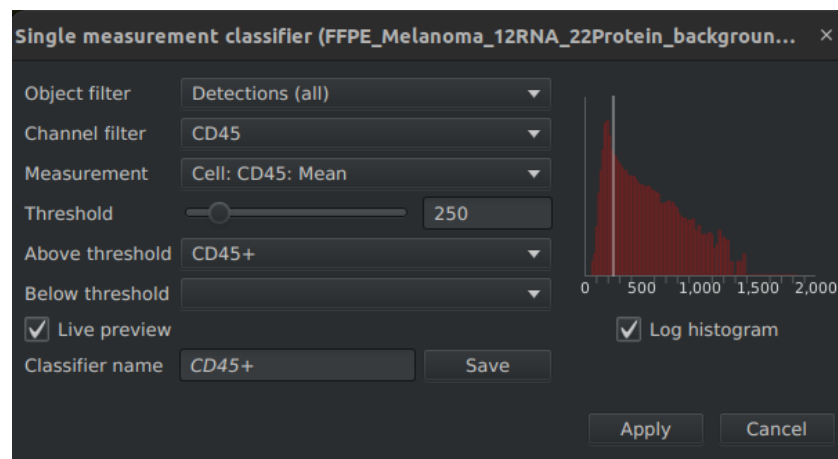


Figure 21. CD45 threshold preview used to separate signal from background.

- The classifier will assign the **CD45+** class to cells whose selected CD45 intensity measurement (**Cell: CD45 mean**) exceeds the chosen threshold (*above 250*). (Figure 21)

Note: You will select the most relevant cell compartment to measure intensity depending on marker used (if marker is cytoplasmatic, nuclear, membrane....).

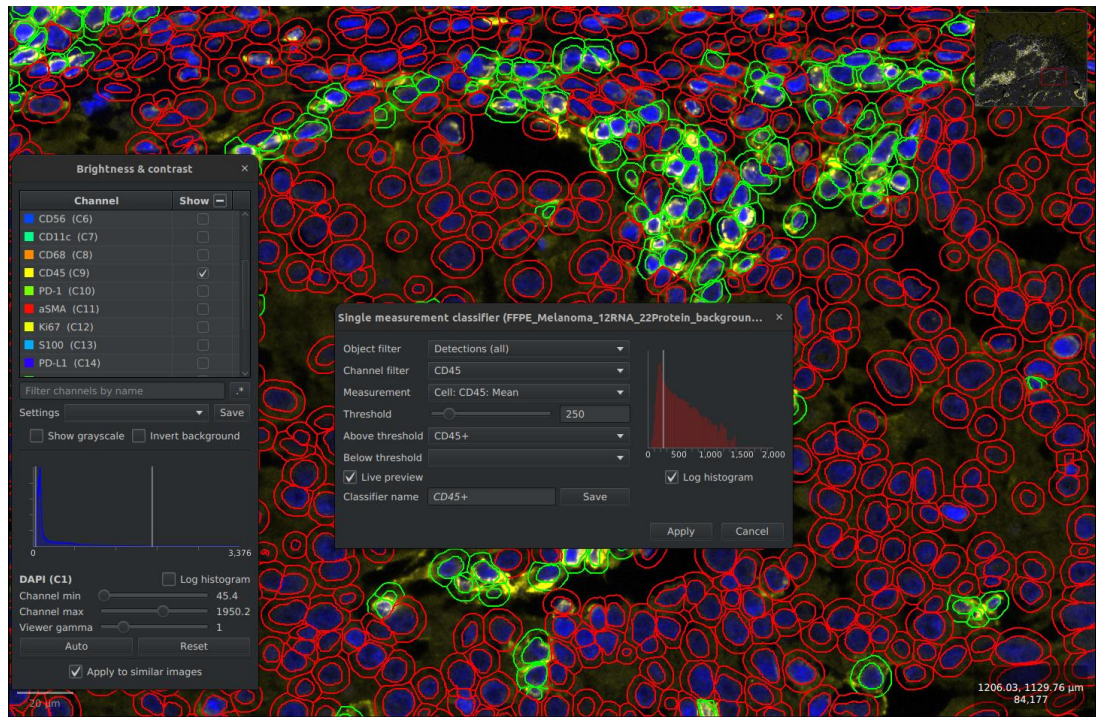


Figure 22. Single-measurement classifier configuration used to assign marker-positive cells from the selected mean-intensity measurement and threshold.

- You can double-click any detected cell and inspect its measurements in the lower part of the **Annotations** panel (Figure 23). This is useful for classification.

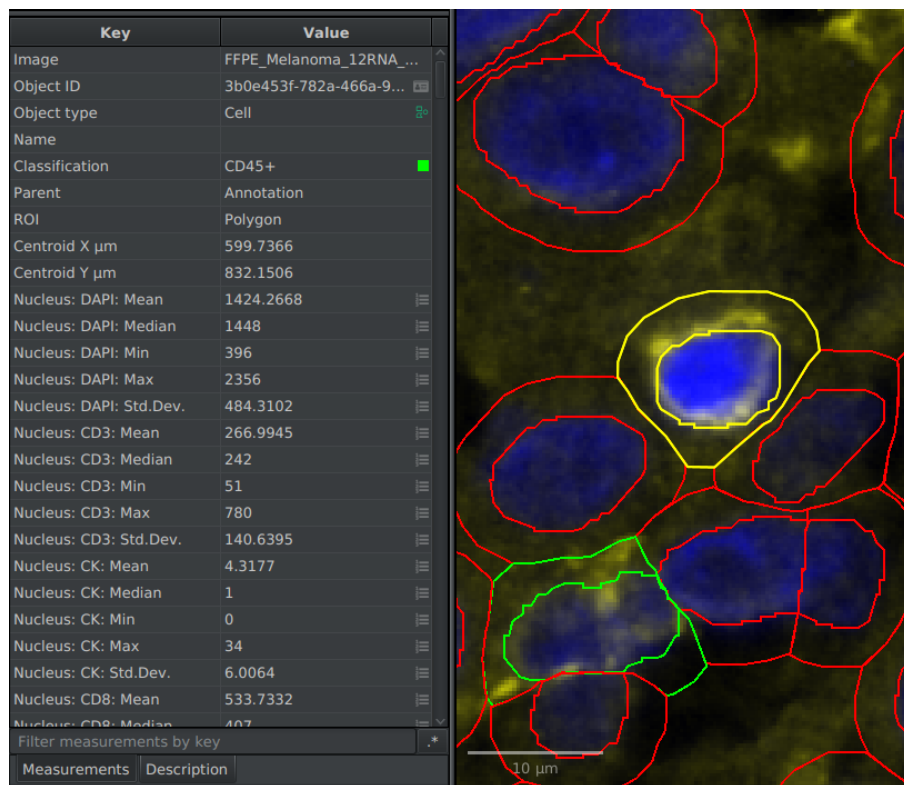


Figure 23. You can double-click any detected cell and inspect its intensity measurements for all cell compartments.

Create CD3+ Classifier

- Activate only the **CD3** channel and adjust the visualization in **Brightness & Contrast**.
- Repeat the same process used for **CD45+ classifier**:
 - Go to **Classify > Object classification > Create single measurement classifier**.
- Configure the classifier using the values shown in the exercise figure for **CD3+ (Figure 23)**.

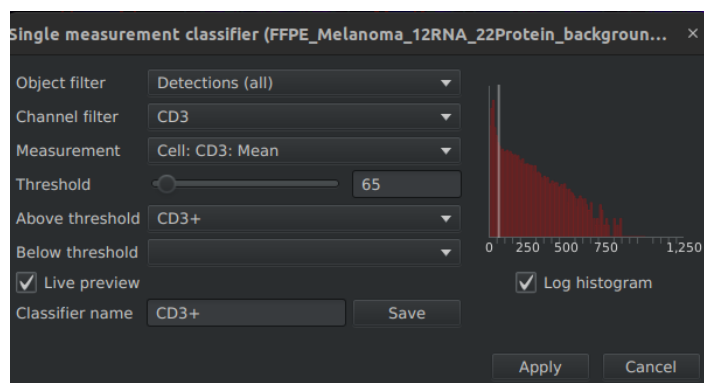


Figure 24. CD3 threshold preview used to separate signal from background

- Then click on **Save** to save the classifier as **CD3+**.
- The classifier will assign the **CD3+** class to cells whose CD3 intensity measurement (**Cell: CD3 mean**) exceeds the chosen threshold (*above 65*). (**Figure 24**)

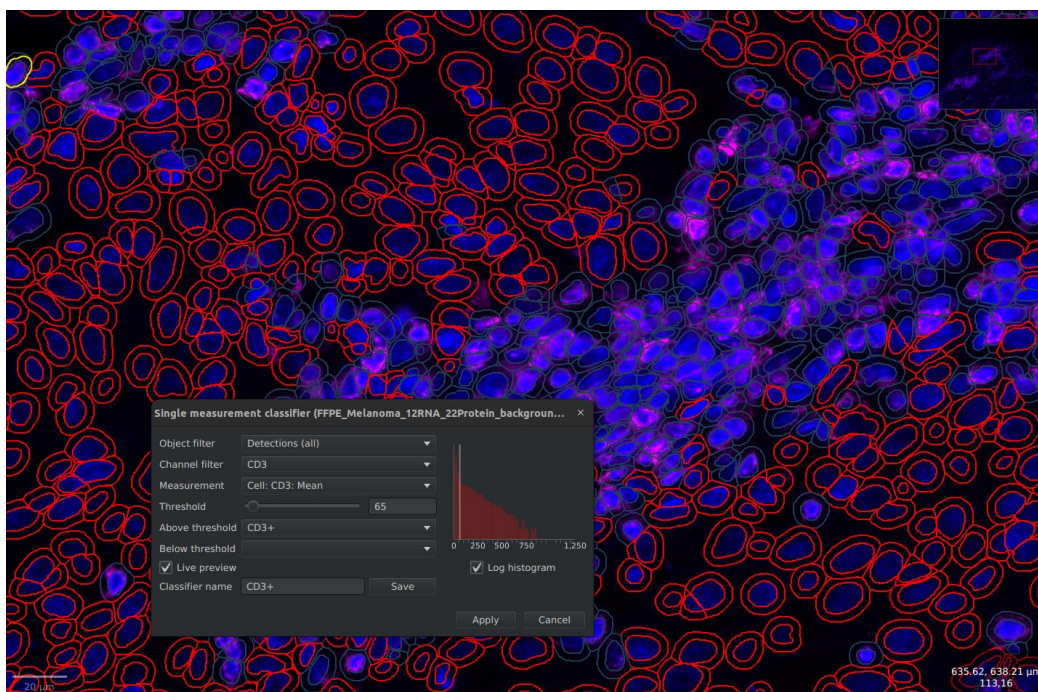


Figure 25. CD3 single-measurement classifier settings used to define CD3-positive cells.

Create CD8+ Classifier

- Activate only the **CD8** channel and adjust the visualization in **Brightness & Contrast**.

- Repeat the same process used for **CD45+** and **CD3+** classifiers:
 - Go to **Classify > Object classification > Create single measurement classifier**.
- Configure the classifier using the values shown in the exercise figure for **CD8+** (**Figure 25**).

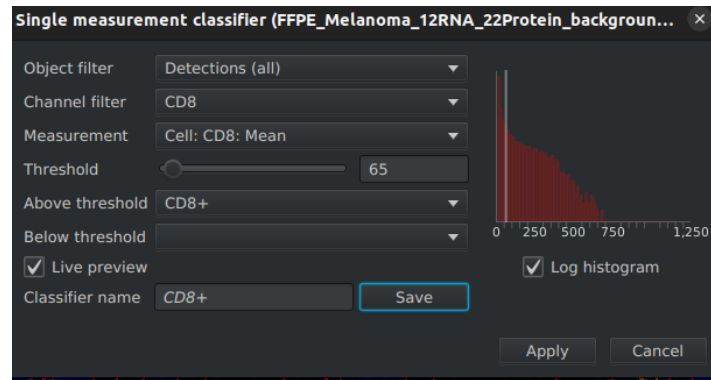


Figure 26. Configure the classifier using the values shown in the exercise for CD8+.

- Then click **Save** to save the classifier as **CD8+**.
- The classifier will assign the **CD8+** class to cells whose selected CD8 intensity measurement (**Cell: CD8 mean**) exceeds the chosen threshold (*above 65*). (**Figure 26**)

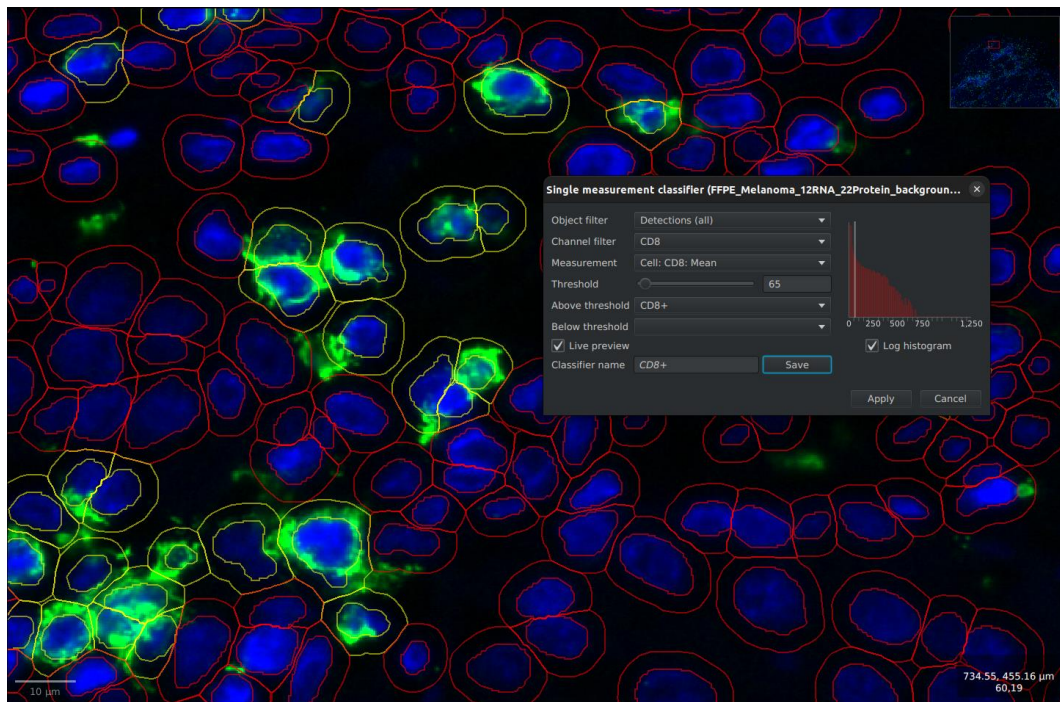


Figure 27. CD8 single-measurement classifier settings used to define CD8-positive cells.

Create CD8- Classifier

- Activate only the **CD8** channel and adjust the visualization in **Brightness & Contrast**.
- Repeat the same process used for **CD8+**:
 - Go to **Classify > Object classification > Create single measurement classifier**.
- Configure the classifier using the values shown in the exercise figure for **CD8-** (**Figure 28**).

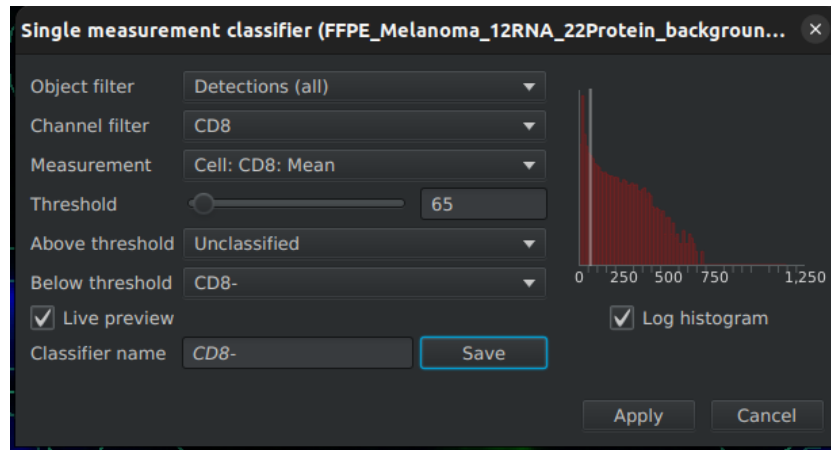


Figure 28. CD8 threshold preview used to separate signal from background.

Important: For the **CD8-** classifier, make sure that **CD8-** is selected for “Below threshold” and “Unclassified” is selected for “Above threshold.” This ensures that only cells with a Cell: CD8 mean value below **65** are classified as CD8-. Cells above the threshold must remain unclassified (red), rather than being assigned to the CD8- class (color defined).

- Then click **Save** to save the classifier as **CD8-**.
- The classifier will assign the **CD8-** class to cells whose selected CD8 intensity measurement (**Cell: CD8 mean**) is below the chosen threshold (*below 65*). (**Figure 28**)

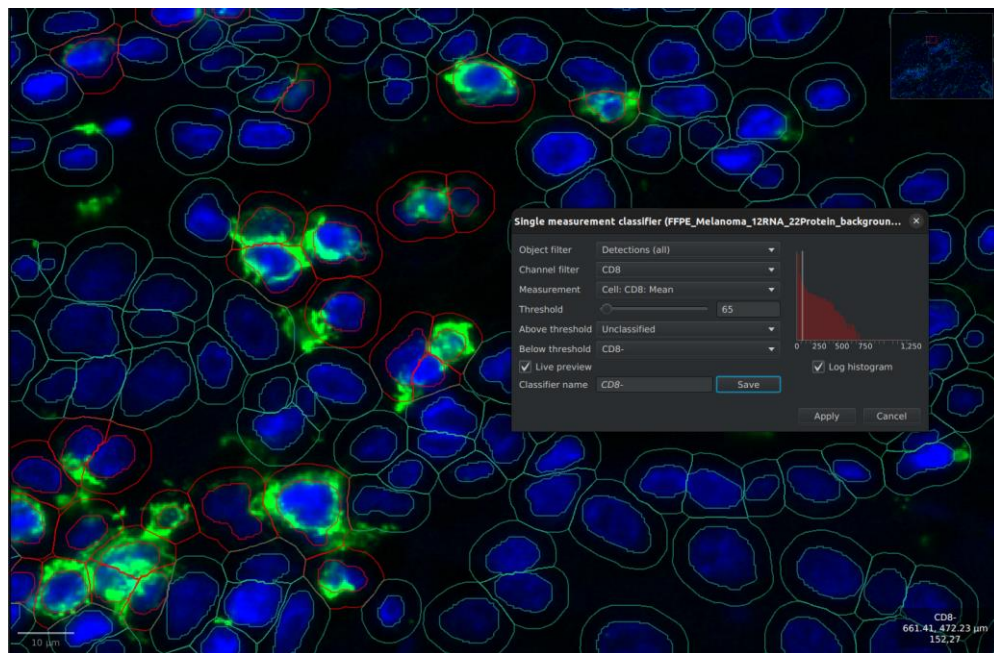


Figure 29. CD8 single-measurement classifier settings used to define CD8-negative cells.

Create a Composite Classifier for TCells (CD45+CD3+CD8-):

- To identify the **T-cell population**, combine the previously created **CD45+**, **CD3+**, and **CD8-** **single** classifiers into a **composite** classifier:
 - Go to **Classify > Object classification > Create composite classifier**.
- Move the corresponding single classifiers (**CD45+**, **CD3+**, **CD8-**) to the right panel using the arrow buttons to create the composite classifier (**TCells**).

- Check that all three classifiers (**CD45+**, **CD3+**, **CD8-**) are included in the composite classifier as **Figure 30**.

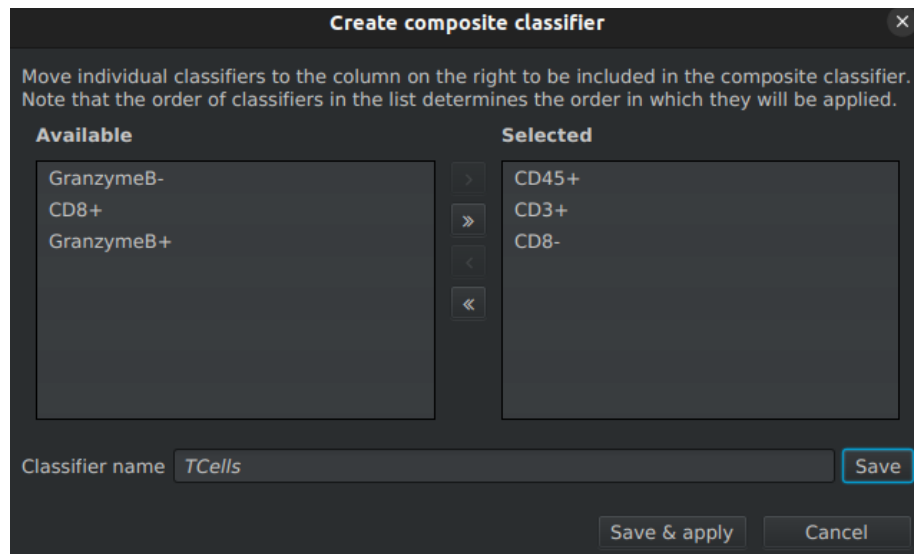


Figure 30. Selection of the CD45+, CD3+, and CD8 single-measurement classifiers used to build a composite T-cell classifier.

- Enter **TCells** as the **Classifier name** and click **Save & Apply** to save the composite classifier and apply it to the cell detections (**Figure 30**)

- After applying the composite classifier, QuPath evaluates each cell independently with the **CD45+**, **CD3+**, and **CD8-** single-measurement classifiers.

Note: The composite classifier does not assign only the **TCells** phenotype. Instead, it combines **all classes assigned to each cell** according to the individual marker criteria. Therefore, a cell that meets only one criterion will receive a **single class**, whereas a cell that meets two or three criteria will receive the corresponding compound classification. For example, a cell classified as both **CD45+** and **CD3+**, but not as **CD8-**, will be assigned the compound class **CD45+:CD3+**. Only cells meeting all three criteria will receive the complete **CD45+:CD3+:CD8-** classification corresponding to the target **TCells** population.

- Depending on the marker-expression profile of each cell, the classifications may include:
 - **Unclassified** (the cell does not meet any of the three classification criteria)
 - **CD45+**
 - **CD3+**
 - **CD8-**
 - **CD3+:CD8-**
 - **CD45+:CD3+**
 - **CD45+:CD8-**
 - **CD45+:CD3+:CD8-** (corresponding to the target **TCells** population)
- To inspect the **resulting phenotypes** at the **single-cell level**, open the **Hierarchy** panel, and review the classification assigned to each cell.

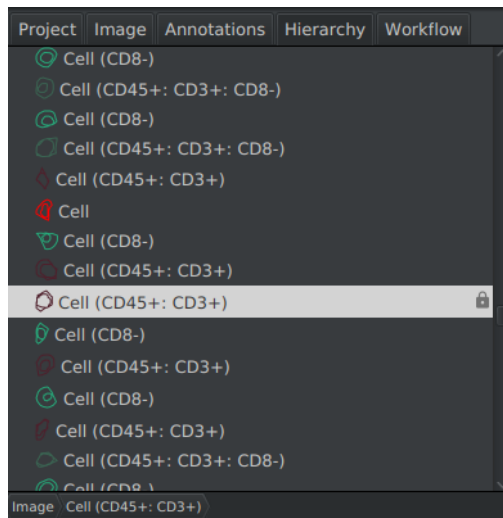
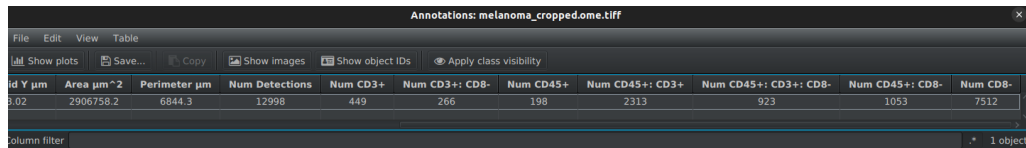


Figure 31. Resulting phenotypes at single-cell level into the Hierarchy panel.

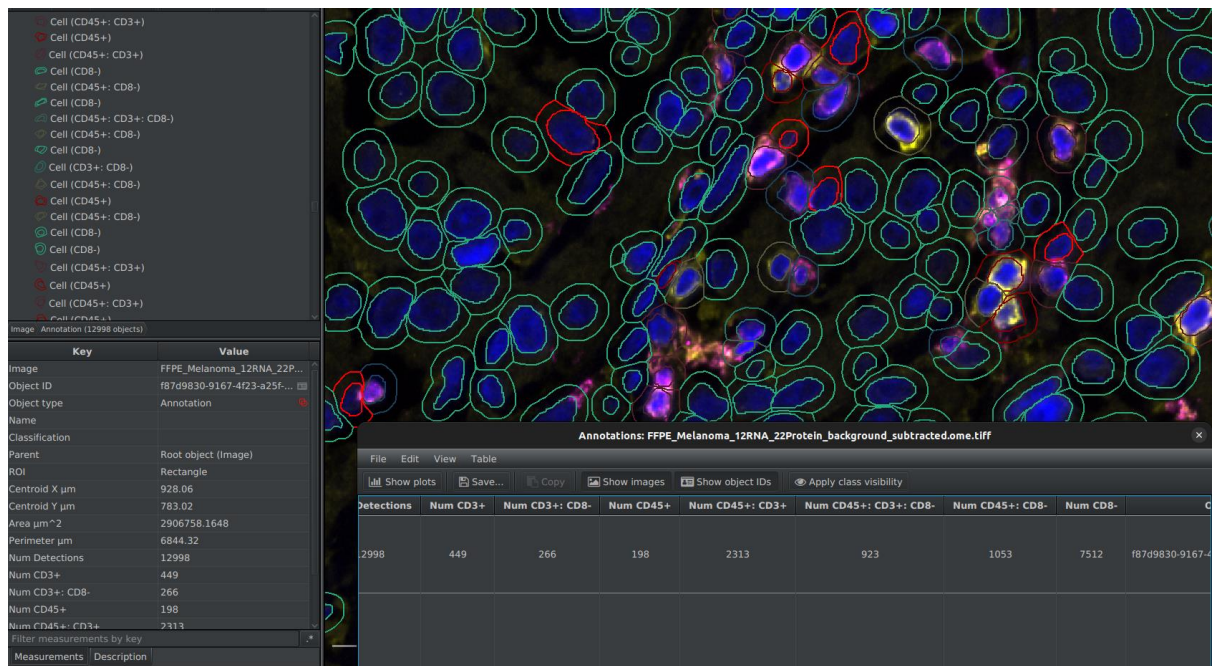
- Compare the classified objects with the fluorescence signals to visually assess the classification.

- Go to **Measure > Show annotation measurements** or  to review the number and proportion of phenotyping cells at annotation level.



id	Y µm	Area µm ²	Perimeter µm	Num Detections	Num CD3+	Num CD3+: CD8-	Num CD45+	Num CD45+: CD3+	Num CD45+: CD3+: CD8-	Num CD45+: CD8-	Num CD8-
3.02		2906758.2	6844.3	12998	449	266	198	2313	923	1053	7512

Figure 32. Phenotyping results for TCells at annotation level.



Classification Results (Left Panel):

- Cell (CD45+, CD3+)
- Cell (CD45+)
- Cell (CD45+, CD3+)
- Cell (CD8-)
- Cell (CD45+, CD8-)
- Cell (CD8-)
- Cell (CD45+, CD3+, CD8-)
- Cell (CD45+, CD8-)
- Cell (CD8-)
- Cell (CD3+, CD8-)
- Cell (CD45+, CD8-)
- Cell (CD45+)
- Cell (CD45+, CD8-)
- Cell (CD8-)
- Cell (CD8-)
- Cell (CD45+, CD3+)
- Cell (CD45+)
- Cell (CD45+, CD3+)

Object Measurements Table (Bottom Panel):

Detections	Num CD3+	Num CD3+: CD8-	Num CD45+	Num CD45+: CD3+	Num CD45+: CD3+: CD8-	Num CD45+: CD8-	Num CD8-
2998	449	266	198	2313	923	1053	7512

Figure 33. General overview of QuPath interface after running TCells classifier.

Create a Composite CD8+TCells (CD45+CD3+CD8+):

- To identify the **CD8+TCells** population, create a **composite** classifier using the previously saved **CD45+**, **CD3+**, and **CD8+** single-measurement classifiers:
 - Go to **Classify > Object classification > Create composite classifier**.
- Move the corresponding single classifiers (**CD45+**, **CD3+**, **CD8+**) to the right panel using the arrow buttons to create the composite classifier (**CD8+TCells**).
- Check that all **three** classifiers are included.
- Enter **CD8+TCells** as the classifier name and click **Save & Apply**

Note: As described for the **TCells** classifier, QuPath evaluates the three criteria independently and generates the single or compound classification corresponding to the marker profile of each cell. Therefore, the output may include **CD45+**, **CD3+**, **CD8+**, and their combinations. Only cells meeting all three criteria will receive the complete **CD45+:CD3+:CD8+** classification, corresponding to the target **CD8+TCells** population.

- Depending on the marker-expression profile, the resulting classifications include:
 - Unclassified
 - CD45+
 - CD3+
 - CD8+
 - CD45+:CD3+
 - CD45+:CD8+
 - CD3+:CD8+
 - CD45+:CD3+:CD8+, corresponding to the target **CD8+TCells** population.
- Inspect the resulting cell classifications in the **Hierarchy** panel.
- Go to **Measure > Show annotation measurements** to review the number and proportion of cells assigned to each phenotype at the annotation level.

Classification	Parent	ROI	Centroid X µm	Centroid Y µm	Area µm ²	Perimeter µm	Num Detections	Num CD3+	Num CD3+: CD8+	Num CD45+	Num CD45+: CD3+	Num CD45+: CD3+: CD8+	Num CD45+: CD8+	Num CD8+
	Root object (Image)	Rectangle	928.06	783.02	2906758.2	6844.3	12998	266	449	1053	923	2313	198	284

Figure 34. Phenotyping results for CD8+TCells at annotation level.

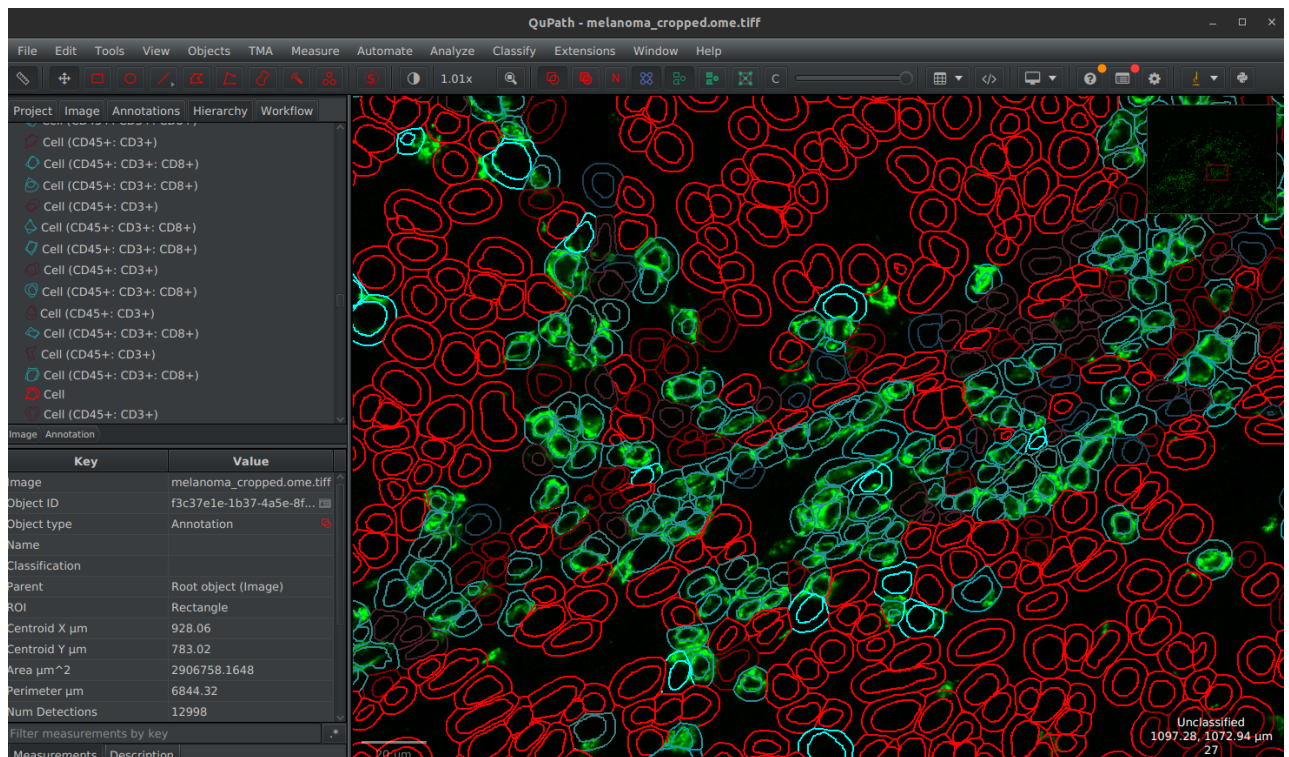


Figure 35. Cell classifications after applying the CD8+TCells classifier, showing single and compound phenotypes.

Reviewing and Editing Saved Classifiers

- Saved object classifiers can be reapplied later from:
 - Classify > Object classification > Load object classifier...**

Caution: Loading and applying a classifier will reclassify the current objects, potentially overwriting previous assignments.

- If you want to inspect or modify the thresholds stored inside a saved classifier at code level:
 - Navigate to: `/cw6_spaom_2026/cw6_qupath/classifiers/object_classifiers`

Important: Composite classifiers are not dynamically linked to the individual classifiers from which they were created. Therefore, if you need to edit a composite classifier, you must load and edit the **composite classifier itself**, not the original single-measurement classifiers.

4. Cell Phenotyping Using ML-Supervised Object Classifiers (ML-based Workflow)

The previous workflow used **Single Measurement Classifiers** to determine marker positivity by applying an intensity threshold to one measurement at a time. While this **threshold-based** strategy is **simple** and **fast**, it may be **suboptimal** when:

- Marker intensity varies considerably across the image.,
- Positive and negative populations are not clearly separated.
- Marker expression overlaps with tissue autofluorescence or background signal.
- Cell populations display heterogeneous expression patterns.
- Phenotype identification requires the simultaneous evaluation of several markers.

A more flexible strategy is to **train a supervised object classifier** using **manually labeled** example cells, instead of relying on a single intensity threshold. This approach allows QuPath to classify the **same populations (T Cells and CD8+ T Cells)** using a **combination of measurements** rather than a single threshold.

Prepare the Training Classes

In this exercise, the supervised classifier will be trained to identify the same populations examined in the threshold-based workflow.

Phenotype	Marker Combination
T Cells	CD45+ CD3+CD8-
CD8+ T Cells	CD45+ CD3+ CD8+

Important: These biological populations are hierarchically related. CD8+ T cells are a subset of the total T-cell population.

Before generating the training examples, make sure that the following classes are available in the **Class list** panel:

- **TCells**
- **CD8+TCells**
- **BG**

Important: The **BG** class will be used for **non-target cells**, poorly segmented objects, **artefacts**, and objects displaying non-specific or **background fluorescence**.

If any of these classes are not available in the **Class list** panel:

1. Locate the **Class list** panel on the left side of the interface.
2. Click on the **+** button to add a new class to the list.
3. Enter the name of the new class, for example, **TCells**.

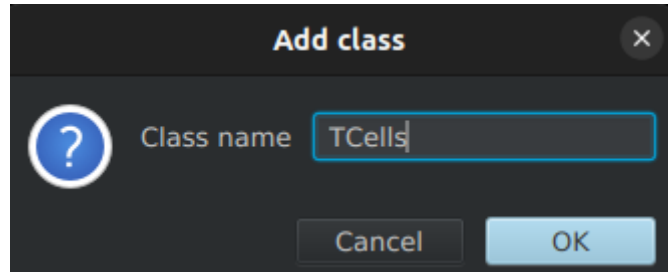
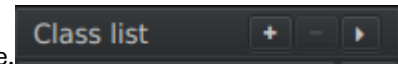


Figure 36. Enter the name of the new class, for example, TCells.

4. Click **OK** to add the class.
5. Repeat the process for all the remaining classes (**CD8+ T Cells**).

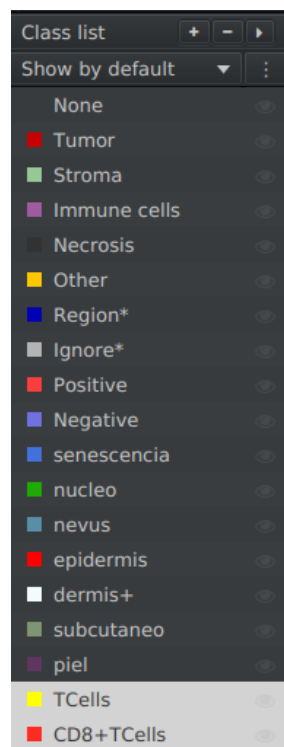


Figure 37. Classes created in the Annotations panel.

Generate a Representative Training Set

Use the cell detections generated by **Cellpose** in **melanoma_cropped.tiff** as candidate training objects for the supervised ML classifier. The classifications obtained with the single-measurement and

composite classifiers can be used as a visual reference to identify representative cells and create the training set.

- Open the **Annotations** panel and check whether a **Full Image Annotation** (Annotation) is present.

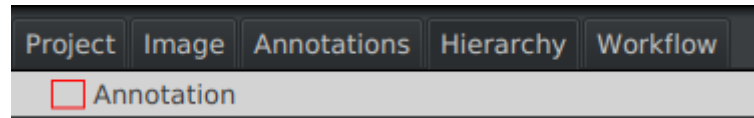


Figure 38. Location of Full Image Annotation.

- If it is present, **select** and **delete** it.

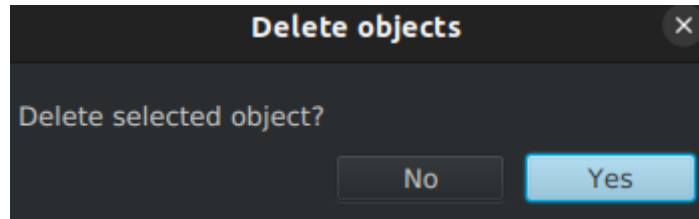


Figure 39. Dialog to delete Full Image Annotation

- Activate the relevant **channels** to identify the phenotype of interest:
 - CD45, C9**
 - CD3, C2.**
 - CD8, C4**
 - DAPI, C1**
- Go to **Measure > Show detection measurements** to open the detection table.
- Locate the **Classification** column.
- Sort the table by clicking the **Classification** column header.

	Object type	Name	Classification	Parent	ROI	Centroid X μm	Centroid Y μm
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	517.75	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	417.36	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	391.31	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	483.45	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	574.43	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	587.28	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	506.78	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	566.06	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	494.31	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	385.89	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	414.29	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	445.45	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	439.12	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	452.13	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	427.54	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	526.62	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	408.47	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	492.93	
e.tiff	Cell		CD45+: CD3+: CD8-	Root object (Image)	Polygon	574.79	

Figure 40. Classification column showing TCells class.

- Find the compound classification corresponding to the population that will be labelled:
 - TCells**
 - Select representative **CD45+ CD3+ CD8-** cells.
 - CD8+TCells**
 - Select representative **CD45+ CD3+ CD8+** cells.
 - BG**
 - Select **non-target cells**, autofluorescent cells, poorly segmented detections, and objects displaying non-specific or background signal.
- Start by generating training examples for the **TCells** class, in the **Annotations** tab, select the **TCells** class:

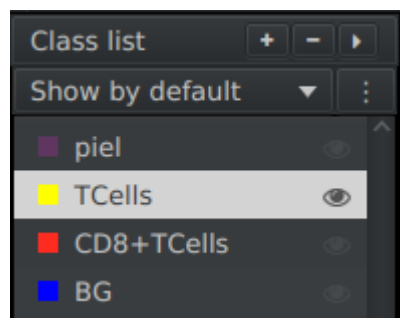
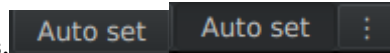


Figure 41. Selection of TCells class in the Annotations panel.

- Enable **Auto set** so that each newly selected training object is automatically assigned to the **TCells** class.



Note: You can use **Auto set** in **Annotations** tab to automatically set all new annotations to the selected class.

- In the detection table, locate cells classified as **CD45+:CD3+:CD8-**
- Use the annotation brush  to select representative cells with the phenotype **CD45+ CD3+ CD8- (TCells)**
- Use the **classifications** generated previously with the **single-measurement** and **composite classifiers** as a guide to locate candidate **TCells**.
- Once a representative set of **TCells** has been labelled, continue with the **CD8+TCells** and **BG** class:

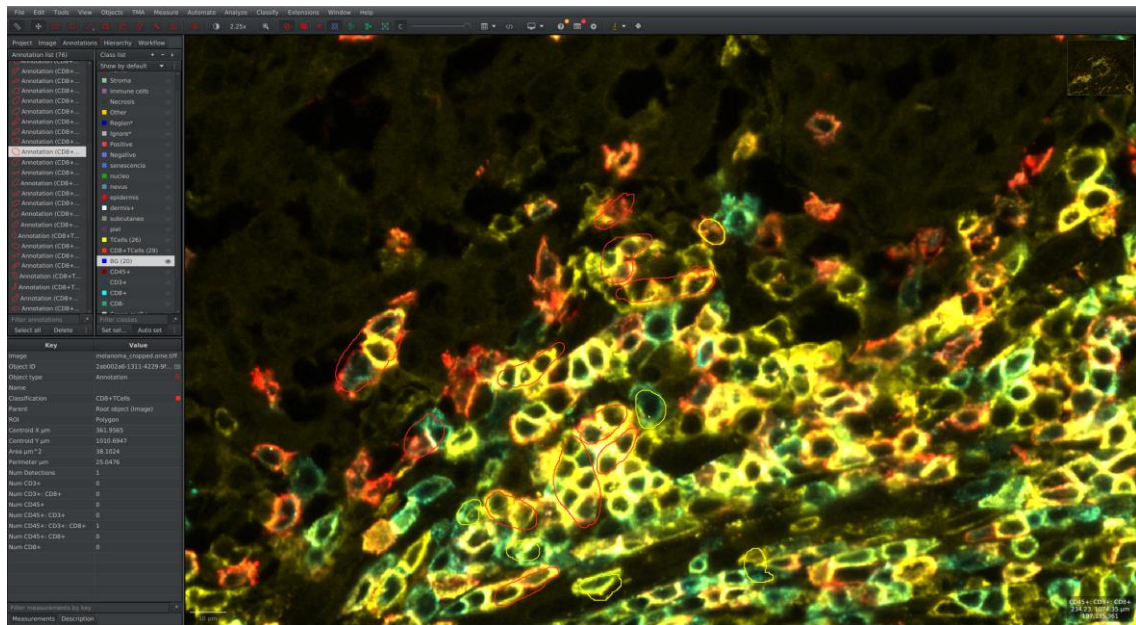


Figure 42. Representative manually labelled cells for CD8+TCells and TCells used to train the supervised object classifier.

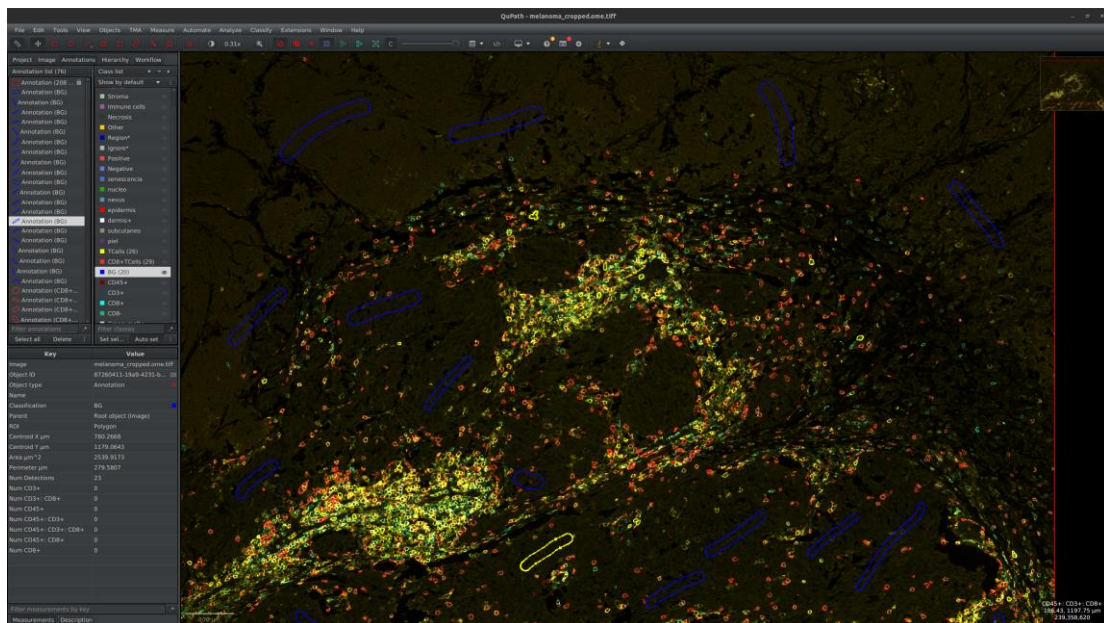


Figure 43. Representative non-target cells and non-specific signal assigned to the BG training class.

Note: You can use the predefined training annotations provided in the path: `/cw6_spaom_2026/cw6_qupath/geojson_masks`

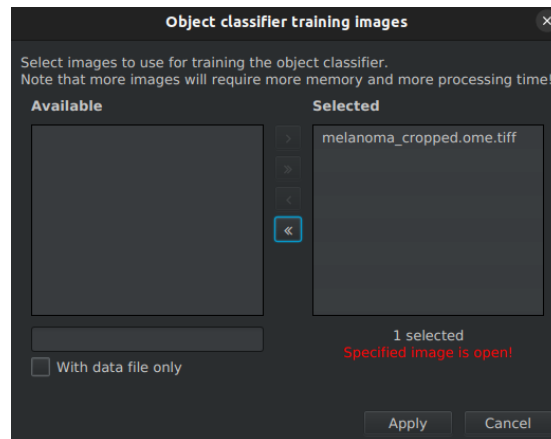
- TCells training annotations: `training_annotation_TCells_melanoma.geojson`
- CD8+TCells training annotations: `training_annotation_CD8+TCells_melanoma.geojson`
- BG training annotations: `training_annotation_BG_melanoma.geojson`

Train the Object Classifier

- Go to **Classify > Object classification > Train object classifier**

In the **training window**:

- Confirm that the **manually labeled cell objects** are available as the training set. And set the following parameters for training:
 - **Object filter:** Detections (all)
 - **Classifier:** Try Random trees (**RTrees**) or Artificial neural network (**ANN_MLP**), both are good choices.
 - **Features:** All measurements
 - **Classes:** All classes
 - **Load training:** `melanoma_cropped.ome.tif`



- Click on **Live update** to toggle whether to calculate classification 'live' while viewing the image.

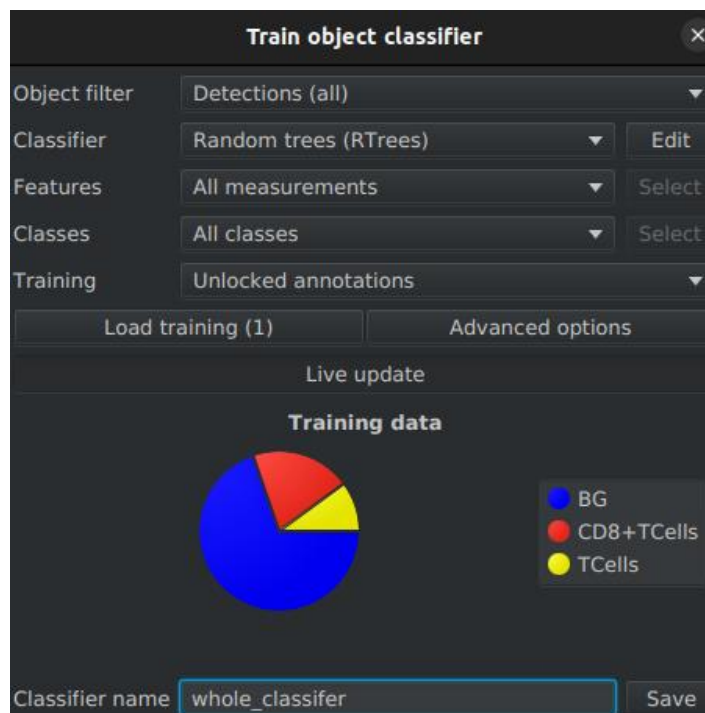


Figure 44. Configuration of the supervised object-classifier training window and representative training examples.

Recommended feature groups: While **Live update** button is pressed, you can add as many label for each class as you desire in order to update the classifier dynamically.

Evaluate the Classification Preview

After training, QuPath will display the **predicted class assignments**. **Review** the results carefully:

- verify that clearly **positive CD8+TCells** cells are classified correctly,
- verify that clearly **positive TCells** cells are classified correctly, same with **BG** cells
- **If needed:**
 - **relabel** misclassified training examples,
 - add more examples from underrepresented classes,
 - **retrain the model**.

Note: This iterative process is often necessary to obtain a robust classifier.

Save and Apply the Trained Classifier

- Once the model performs satisfactorily, save it with a descriptive name as: **whole_classifier**.

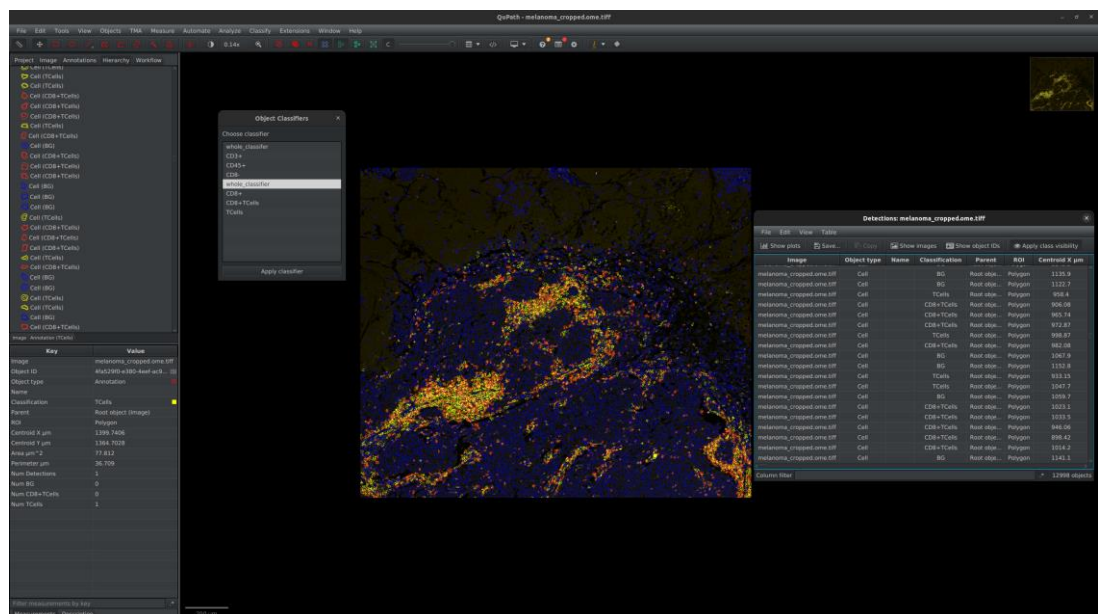
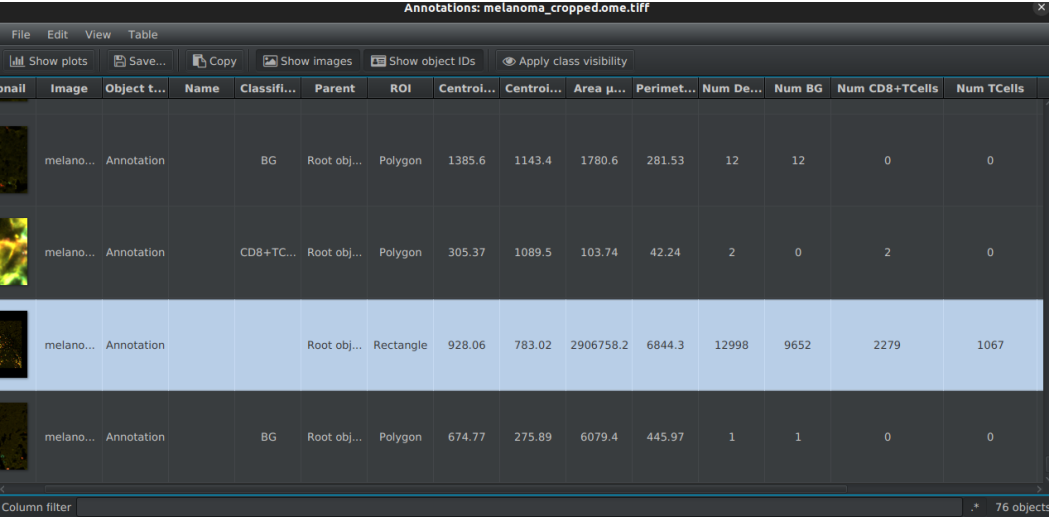


Figure 45. Predicted TCells, CD8+TCells, and BG classes after applying the trained object classifier.

- At this point, the classifier should assign the populations as the threshold-based strategy, but using a **ML-supervised model** rather than a single intensity threshold.
- Go to **Measure > Show annotation measurements** to check N of cells belonging to each population (**TCells**, **CD8+TCells** and **BG**) within the **full image annotation**.



Thumbnail	Image	Object t...	Name	Classifi...	Parent	ROI	Centroi...	Centroi...	Area μ...	Perimet...	Num De...	Num BG	Num CD8+TCCells	Num TCells
	melano...	Annotation		BG	Root obj...	Polygon	1385.6	1143.4	1780.6	281.53	12	12	0	0
	melano...	Annotation		CD8+TC...	Root obj...	Polygon	305.37	1089.5	103.74	42.24	2	0	2	0
	melano...	Annotation			Root obj...	Rectangle	928.06	783.02	2906758.2	6844.3	12998	9652	2279	1067
	melano...	Annotation		BG	Root obj...	Polygon	674.77	275.89	6079.4	445.97	1	1	0	0

Figure 46. Quantification of relevant phenotypes for the full image annotation.

- Saved object classifiers can be reapplied later from:
 - **Classify > Object classification > Load object classifier**

Caution: Loading and applying a classifier will reclassify the current objects, potentially overwriting previous assignments.

- If you want to **inspect** or **modify** the saved classifier:
 - Navigate to: `/cw6_spaom_2026/cw6_qupath/classifiers/object_classifiers`

5. Tumor Detection using a Threshold-based Pixel Classifier (Threshold-based Workflow)

In this section, the **tumor compartment** will be identified from the **S100** signal using a **threshold-based pixel classifier**. Pixels with a S100 intensity **above** the selected threshold will be assigned to the **Tumor class** and subsequently, the resulting **Tumor mask** will be converted into a **Tumor annotation**.

- Reuse `melanoma_cropped.ome.tiff` image.
- Hide the **training annotations** and classified **cell detections** (**TCells** and **CD8+TCCells**) from the previous exercise by using the visibility controls in the QuPath toolbar:



Note: Hiding annotations or detections only changes their visibility in the viewer. The objects, classifications, and measurements are not deleted or modified

- Open **Brightness & Contrast** and activate only **S100(C13)** channel.
- Adjust the **S100** display range to visually distinguish **S100-positive tumor regions** from **background** and **non-tumor tissue**, and **non-specific signal**.

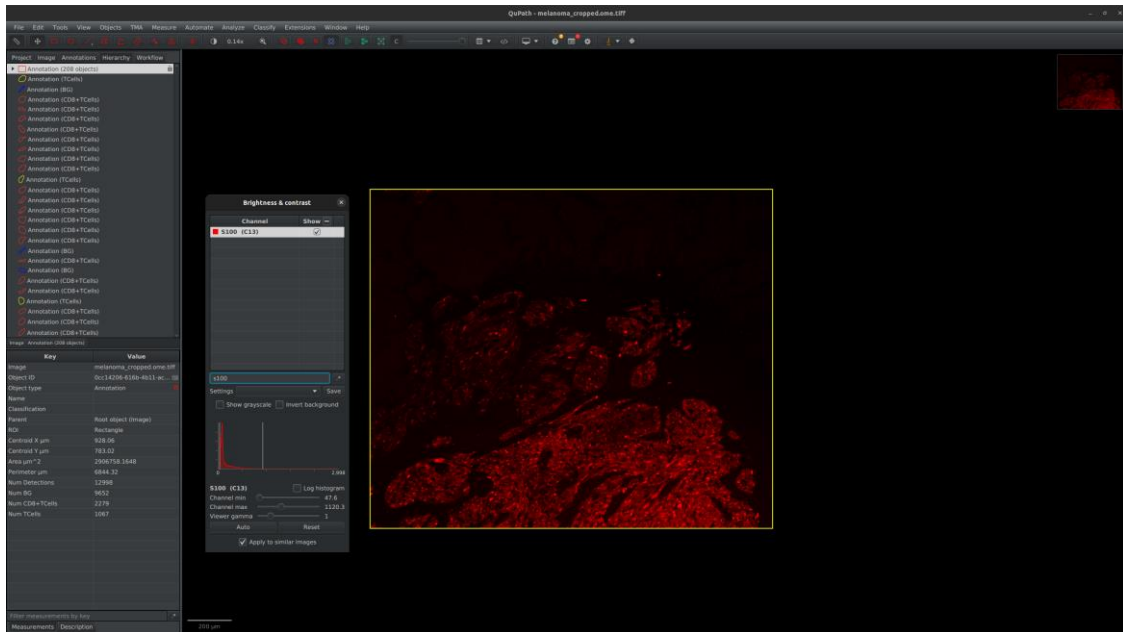
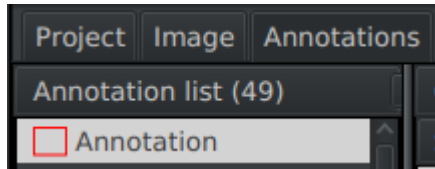


Figure 47. S100 channel display and histogram adjusted to distinguish tumor-associated signal from background before threshold.

- Create a **full image annotation** by **Objects > Annotations... > Create full image annotation**
- Select the **Full Image Annotation** in the **Annotations** panel



Important: The selected Full Image Annotation will be used as the parent region when the thresholded Tumor mask is converted into annotation objects. Make sure that the Full Image Annotation, rather than a TCells or CD8+TCells training annotation, is selected.

- Create the **Threshold-based Pixel Classifier**:
 - Go to **Classify > Pixel classification > Create thresholder** (or **Ctrl+L** and search for **Create thresholder**)
- Configure the **classifier** using the parameters shown in **Figure 48**.
 - **Resolution:** Llow

Note: The **Extremely low** resolution is appropriate for identifying broad tumor compartments rather than fine cellular or subcellular structures. This setting also **reduces processing time**. However, very small tumor regions and fine boundary details may be represented less precisely.

- **Channel:** S100
- **Prefilter:** Gaussian

Note: The **Gaussian prefilter** smooths small pixel-to-pixel intensity variations before thresholding. This can reduce fragmentation of the Tumor mask, although excessive smoothing may merge adjacent regions or remove small S100-positive structures.

- **Smoothing sigma:** 0
- **Threshold:** 180
- **Above Threshold:** Tumor_TH
- **Below Threshold:** Empty
- **Region:** Everywhere
- **Classifier name:** Tumor_TH
- Click **Save** once the preview is satisfactory.

Important: Do not close the **Create Threshold** window after saving the classifier. The thresholded mask will be converted into Tumor annotation in next step.

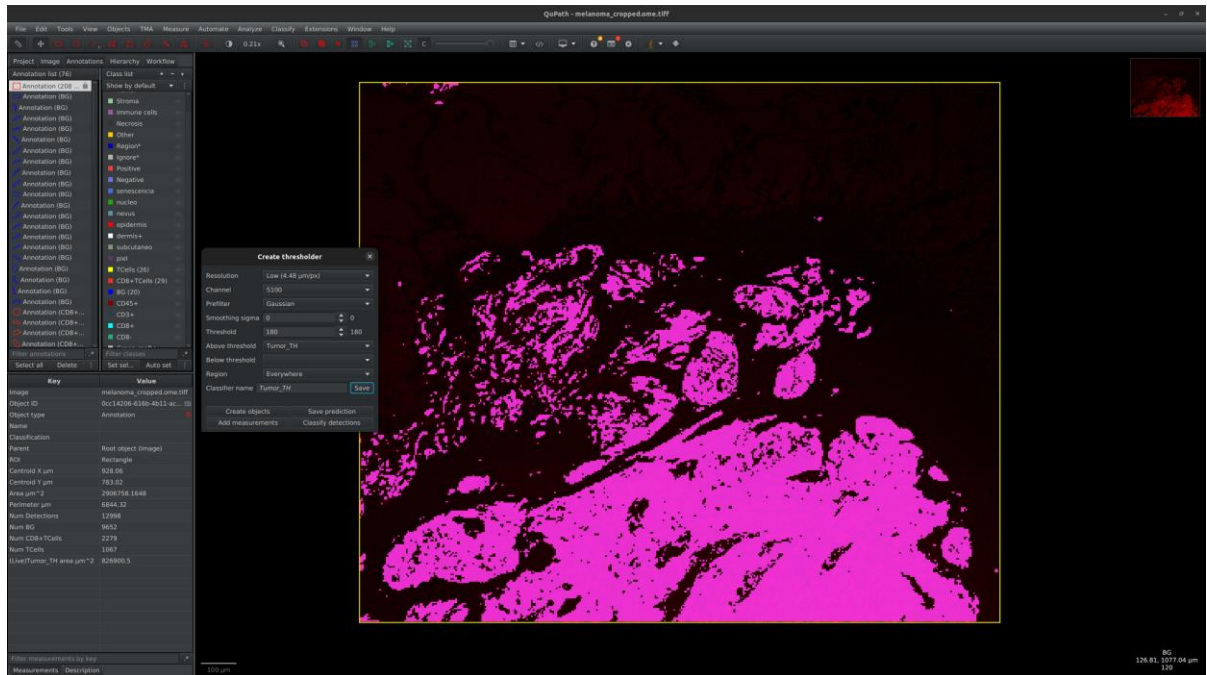


Figure 48. Threshold-based pixel-classifier configuration for detection of S100-positive tumor regions.

Note: Using a threshold-based approach does not give the most accurate results since it is only based on a single intensity measurement that does not catch the whole variability of the tumor signal. Let's see if we can improve the results with the ML approach.

Create Tumor Objects

The pixel classifier initially generates a **mask**. To use the tumor compartment in subsequent spatial analyses, the **Tumor mask** must be converted into a **Tumor annotation**.

- Confirm that the **Full Image Annotation** is selected.
- Click **Create objects** in the **Create threshold** window.
- When prompted to **Choose the parent objects**, select: **Current selection** and click **OK**.

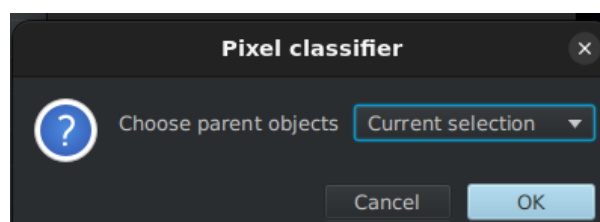


Figure 49. Dialog to Create objects.

Important: Selecting **Current selection** restricts object creation to the selected parent annotation. If a TCells or CD8+TCells training annotation is selected instead of the **Full Image Annotation**, Tumor objects may be generated only within that small region.

- Configure the parameters in the **Create objects** window according to **Figure 50**.

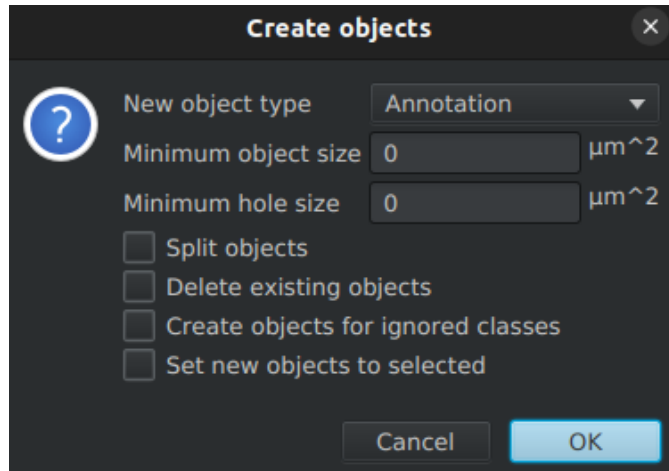


Figure 50. Object-creation parameters used to convert the tumor pixel mask into Tumor annotation.

- Once the parameters are defined, click **OK** and close the **Create Threshold** window.
- QuPath will generate a **Tumor annotation** corresponding to the **S100-positive signal**.
- Go to **Hierarchy** panel and observe how **Tumor annotation (Annotation (Tumor))** has been created under **Full image Annotation**.

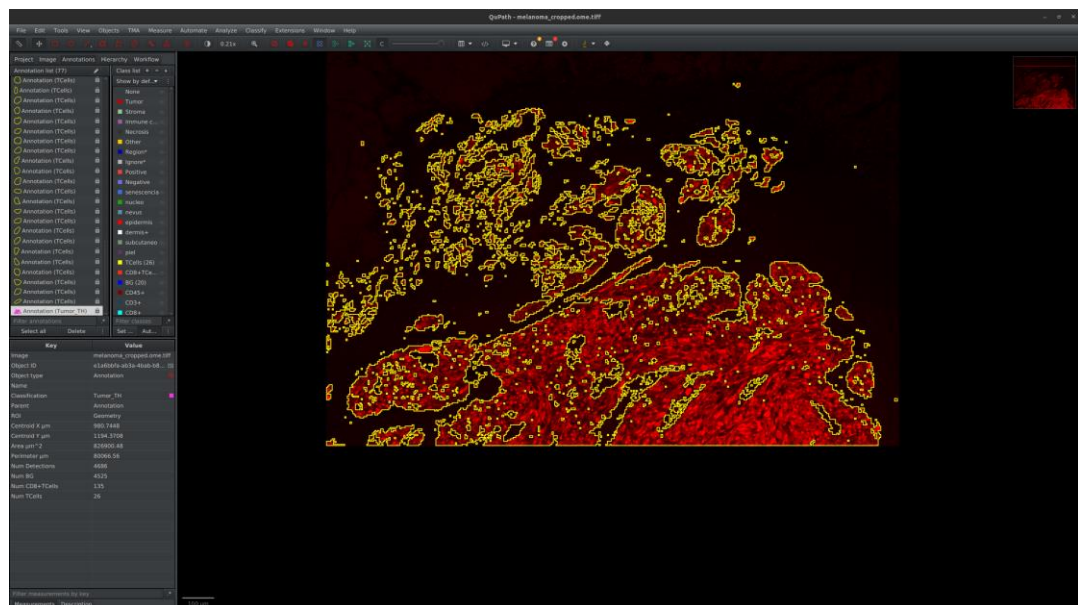
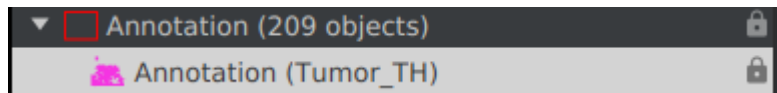


Figure 51. Tumor annotation generated from the S100 threshold-based pixel classifier.

Note: You can use the predefined S100 annotation provided in the path: `/cw6_spaom_2026/cw6_qupath/geojson_masks`

- Tumor_TH annotation:** `annotation_Tumor_TH_melanoma.geojson`

6. Tumor Detection using a Trained Classifier (ML-based Workflow)

In this section, the tumor compartment will be identified using a **supervised pixel classifier** trained with manually annotated examples. Unlike the **threshold-based approach**, which relies on a **single S100 intensity threshold**, the trained classifier learns to distinguish between:

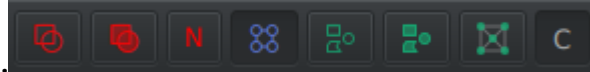
- **Tumor_ML**: pixels representing tumor tissue.
- **BG**: pixels representing non-tumor tissue, background, artefacts, or other structures that should not be included in the tumor compartment.

Note: A **trained pixel classifier** is **not** automatically **better** than a **threshold-based classifier**. Its performance depends on the quality, accuracy, and balance of the training annotations.

Note: Include both strong and weaker tumor regions, as well as challenging borderline regions. This will help the classifier learn the variability present in the image.

Prepare the Image and Viewer

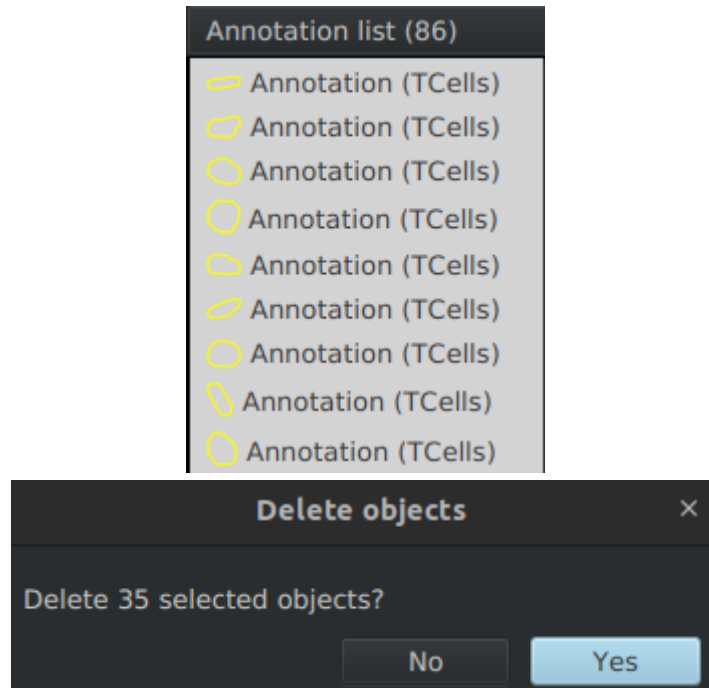
- Reuse **melanoma_cropped.tiff** image.
- Make sure that the **S100 channel** is visible, since S100 will provide information for identifying tumor regions.
- Temporally hide **cell detections/classifications** and **training annotations (TCells and CD8+TCells)**.



Prepare the Annotations used for Training

The **Train pixel classifier** window automatically uses the classified training annotations available in the loaded image. Temporarily remove the **manually** drawn training annotations assigned to:

- **TCells**
- **CD8+TCells**
- **BG**, if **BG_ML** will be used as the single background class
- Any other class unrelated to tumor pixel classification
- Select the annotations in **Annotation list** panel (**TCells**, **CD8+TCells**, **BG** and **Tumor_TH**), then click on **supr** button on keyboard and on **Delete objects** dialog click on **Yes** :



Caution: Remove **only** the manually drawn training annotations. Do not delete the **Cellpose** detections or their cell classifications.

Create the Pixel-Classification Training Classes

- Go to the **Class list** in the **Annotation** panel.
- Click on the **+** next to the **Class list** panel.
- A dialog named **Add class** will pop-up, in **Class name** type:
 - Tumor_ML**
 - BG_ML**

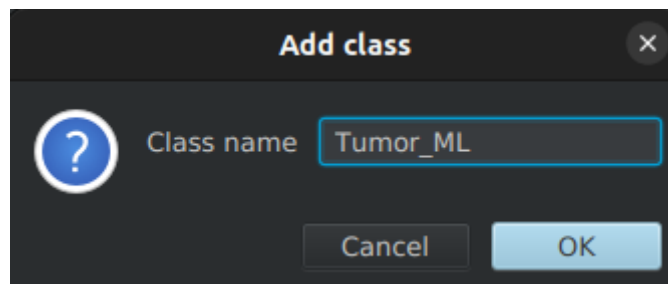


Figure 52. Dialog to Add class in QuPath.

Important: In this exercise, the name **Tumor_ML** is used to distinguish the tumor compartment generated with the **supervised ML pixel classifier** from the **Tumor class** generated with the **threshold-based pixel classifier**.

Drawing the Training Annotations

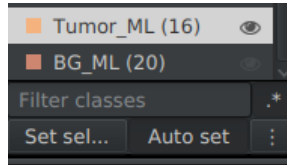
- First, use **Auto set** tool in **Annotations** panel to assign class for each annotation automatically, while drawing new annotations.

Note: With **Auto set** enabled, each newly drawn annotation is **automatically** assigned to the **class** currently selected in the **Class list**. Always check the selected class before drawing new training regions.

- Use the **Brush** tool  to draw representative regions for both **Tumor_ML** and **BG_ML**.

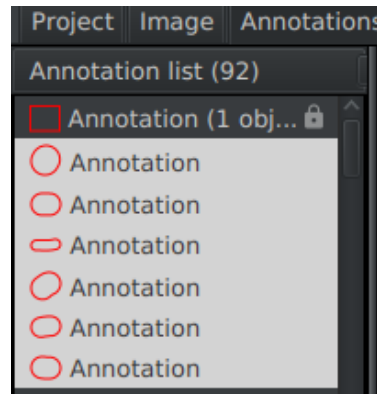
Note: Training with huge areas is discouraged because the model becomes harder to refine, and class imbalance can become problematic.

- Create 10-20 **annotations** for each **Tumor_ML/BG_ML class** from **different parts** of the image considering **variability** (morphology, intensity...).

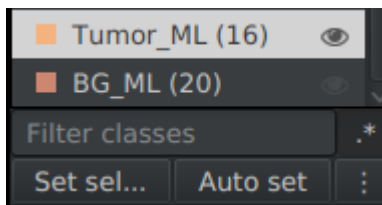


Assign Classes Manually (if Auto Set Was Not Used)

- Select the annotations in the image or **Annotations** panel.



- Select the appropriate class in the **Class list** (**Tumor_ML**, or **BG_ML**)
- Click **Set selected**.



- Confirm that the annotation has received the correct class by activating .

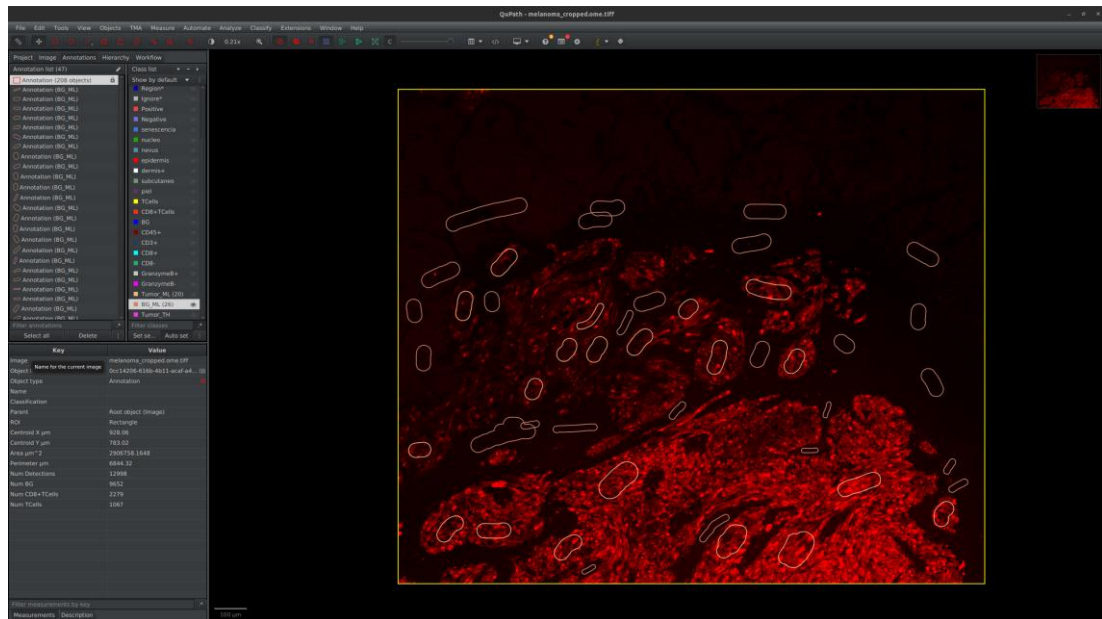


Figure 53. Representative *TumorML* and *BG_ML* training annotations used for supervised pixel classification.

Note: You can use the predefined **BG_ML** and **Tumor_ML** annotations provided in the path:
/cw6_spaom_2026/cw6_qupath/geojson_masks

- **Tumor_ML** annotation: `training_annotation_Tumor_ML_melanoma.geojson`
- **BG_ML** annotation: `training_annotation_BG_ML_melanoma.geojson`

Train the Pixel Classifier

- Before training, please make sure you have the following configuration to hide cell detections and just show training masks.



- Go to **Classify > Pixel classification > Train pixel classifier**
- Configure the classifier using the parameters shown in **Figure 54**:

- **Classifier:** Random Trees (RTrees)

Note: **RTrees** is a suitable starting point for this exercise. Other classifiers, such as **ANN_MLP**, can also be explored, but their results may depend more strongly on the selected features and training set.

- **Resolution:** Very low

Note: The **Very low** resolution is appropriate for identifying broad tumor compartments and reduces processing time. However, very small tumor regions and fine boundary details may be represented less precisely at this resolution.

- **Region:** Everywhere
- **Load training:** add `melanoma_cropped.ome.tiff` and click **Apply**.

- Click **Live prediction**
 - Inspect the predicted classification overlay across multiple areas of the image.

Note: Use **Live prediction** to see how the classifier responds and then refining the training set iteratively by adding more annotations.

- Use the **C** key to hide/show the classification mask to compare with the original image.



- Adjust overlay opacity from the toolbar if necessary

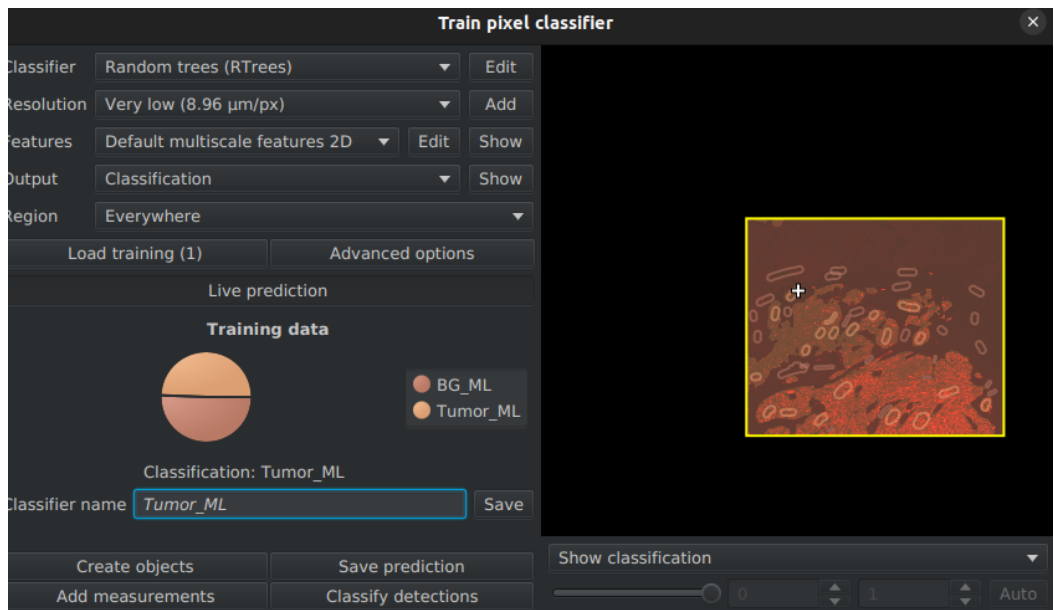


Figure 54. Parameter Configuration used for supervised pixel classification.



Figure 55. Tumor_ML/BG masks after the application of the supervised pixel classifier.

Save the Trained Tumor Pixel Classifier

- Once the prediction is satisfactory, in **Classifier name**, enter **Tumor_ML**
- Then click **Save**



Important: Saving the classifier stores the trained model, but it does not automatically create a Tumor_ML annotation. The predicted pixel mask must be converted into annotation objects in the next step.

Create Tumor Annotation from the Trained Pixel Classifier

After saving the classifier, convert the predicted **Tumor_ML** pixel mask into **annotation**:

- Confirm that the **Full Image Annotation** is selected.
- In **Train pixel classifier** window, click on **Create objects**

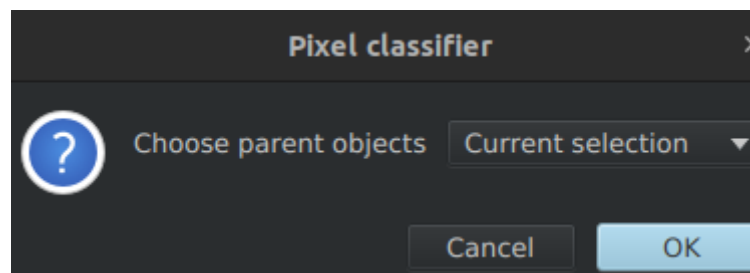


Figure 56. Create objects dialog used to convert a pixel-classification mask into annotations.

- Choose parent objects: **Current selection** and Click **OK**
- Configure the object-creation parameters on **Create objects** window using the same values applied in the threshold-based workflow (**Figure 57**), so that both methods are comparable.

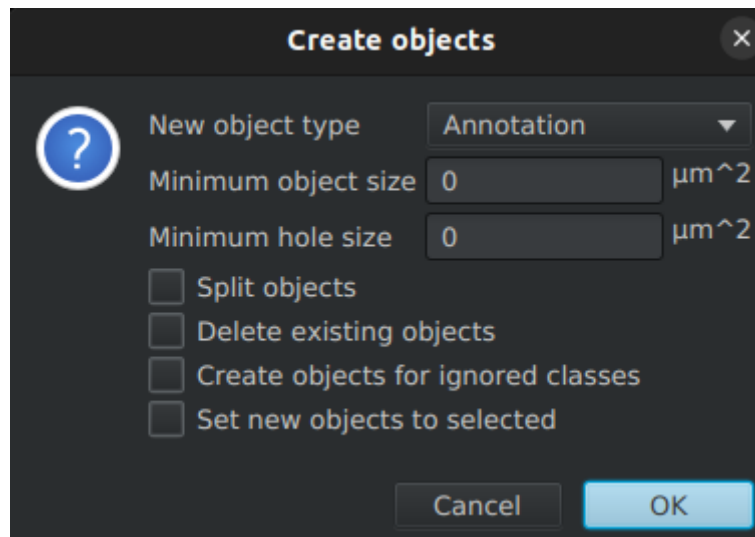
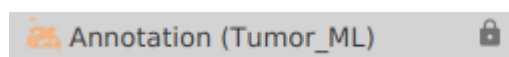


Figure 57. Object-creation settings used to convert the trained pixel-classification mask into annotations.

- Click **OK**
- Close the **Train pixel classifier** window
- Confirm that annotation for **BG_ML** and **Tumor_ML** has been created. They will appear as locked annotations in the **Annotation list**.



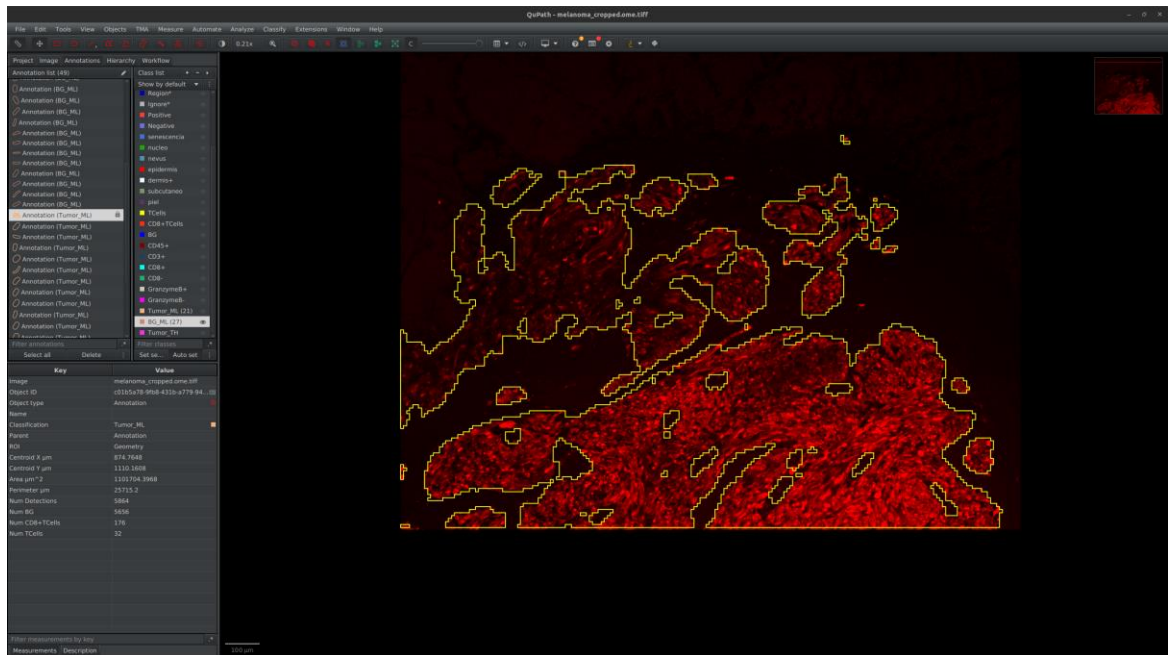


Figure 58. Tumor_ML annotation generated from the trained pixel classifier and displayed in yellow.

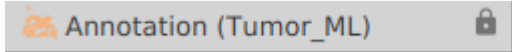
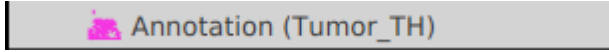
7. Spatial Analysis: Distance to the Tumor Border and Cell-to-Cell Distances

Spatial analysis can be used to investigate how the identified immune-cell populations are distributed in relation to the tumor compartment and to other cells. In this exercise, we will calculate:

- The **distance** from each detected **cell** to the **tumor** border.
- The **distance** from each **cell** to its **nearest neighboring cells**.
- The **proximity** of the **target** immune phenotypes:
 - **TCells:** CD45+:CD3+:CD8–
 - **CD8+TCells:** CD45+:CD3+:CD8+

Prepare the Image for Spatial Analysis

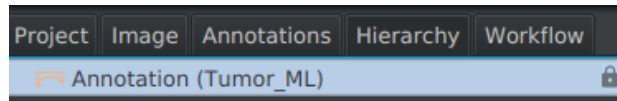
Before starting the spatial analysis, you must have in your QuPath project:

1. A **tumor annotation** generated using either:
 - a. The trained pixel classifier **Tumor_ML**, producing a **Tumor_ML** annotation.
 
 - b. The threshold-based pixel classifier **Tumor_TH**, producing a **Tumor** annotation.
 
2. **Cell detections** generated from the DAPI signal.
3. **Cell classifications** generated with the single and composite classifiers created in the previous sections:
 - a. **CD45+:CD3+:CD8–**, corresponding to **TCells**.
 - b. **CD45+:CD3+:CD8+**, corresponding to **CD8+TCells**.

Distance to the Tumor Border

This analysis calculates the **shortest distance** from each **cell** to the **boundary** of the selected **tumor annotation**. This is useful for distinguishing **cells** that are:

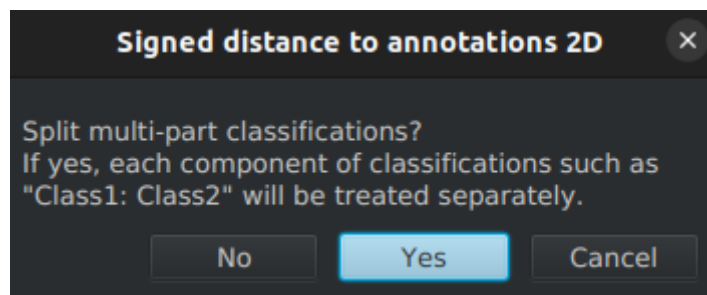
- I. **inside the tumor compartment,**
 - II. **at the invasive margin,**
 - III. **outside the tumor region** in the surrounding microenvironment.
- Open the **Annotations** list on the **Annotations** panel.
 - Select the **Tumor_ML** annotation.



Important: Selecting the wrong annotation may result in distances being calculated from an incorrect reference object. If the tumor annotation consists of several disconnected annotation objects, ensure that all the relevant **Tumor_ML** annotations are selected before running the analysis.

- Go to **Analyze > Spatial analysis > Signed distance to annotation 2D**.

Important: When QuPath asks whether to split multipart classifications, select **No**, because we want to analyze the complete classifications. For example, **CD45+:CD3+:CD8+** should remain a single combined phenotype rather than being divided into separate **CD45+**, **CD3+**, and **CD8+** components.



- Open the **detection measurement table** by **Measure > Show detection measurements** or



Signed distance to annotation Tumor_ML μm
642.97
642.8
637.71
634.88
630.48
622.93

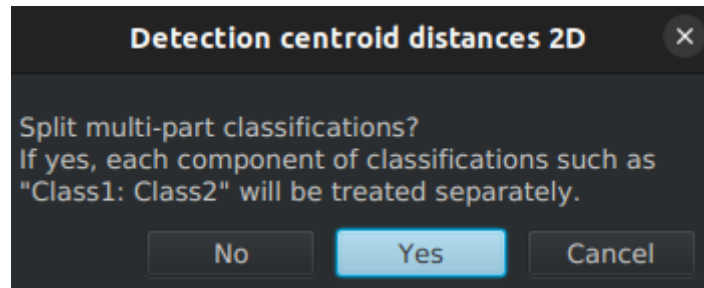
Figure 59. Column added containing the shortest distance from each cell to the boundary of the Tumor and Tumor_ML annotation.

- A new **column** will be added containing the **shortest distance** from each **cell** to the boundary of the selected **Tumor_ML** annotation:
 - **Negative values:** cell centroid is located inside the tumor annotation.
 - **Values close to zero:** cell centroid is located close to the tumor boundary.
 - **Positive values:** cell centroid is located outside the tumor annotation.

Distance to the Nearest Neighbor Cells

Nearest-neighbor analysis measures how **closely** each **cell** is located to **cells** belonging to the same or different **classifications**. This analysis can be used to investigate:

- I. Local clustering of immune cells.
 - II. Proximity between T-cell populations.
 - III. Spatial separation between phenotypes.
 - IV. Whether **CD8+TCells** accumulate near other immune cells.
 - V. Whether **TCells** and **CD8+TCells** show different spatial patterns inside and outside the tumor.
- Go to **Analyze > Spatial analysis > Detection centroid distances 2D**.
 - When asked whether to split multipart classifications, select **No**.



- Wait for QuPath to calculate the measurements.
- Open the measurement table by going to **Measure > Show detection measurements**

ML μm	Distance to detection TCells μm	Distance to detection CD8+TCells μm	Distance to detection BG μm	Object ID
	591.99	543.55	0	1253dd17-8004-467
	586.02	540.77	0	40eb9704-b1a6-4ce
	583.6	536.53	0	f6618631-b006-4e5
	574.86	530.88	0	7169e398-f025-49e:
	567.12	524.7	0	bf4d21ae-9b84-409:

Figure 60. Results of running the shortest among cells from the same or different populations.

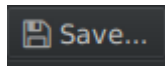
- New **columns** will be added to the measurement table. These columns report the distance from **each cell centroid** to the **nearest cell** belonging to each **population**:
 - The distance from a **TCell** to the nearest **TCell**.
 - The distance from a **TCell** to the nearest **CD8+TCell**.
 - The distance from a **CD8+TCell** to the nearest **CD8+TCell**.
 - The distance from a **CD8+TCell** to the nearest **TCell**.
 - The distance from either target population to other single or compound classifications present in the image.

Important: A shorter distance indicates that cells are located more closely together, but it does not by itself demonstrate a biological interaction. Cell density, tissue architecture, tumor area, and the total number of cells in each population should also be considered.

Export Cell/Annotation Measurement Table

- **Export Cell Detection measurements:** Go to **Measure → Show detection measurements** to export cell-level classifications, marker intensities, coordinates and spatial measurements.

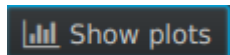
- Export the table as a **.csv** or tab-delimited **.txt** file using the **Save** button



. Use a descriptive filename that identifies the image, analyzed region and classification strategy.

- **Annotation measurements:** Go to **Measure** → **Show annotation measurements** to export region-level information, including annotation names, areas and object counts.
 - Export the table as a **.csv** or tab-delimited **.txt** file using a descriptive filename.

8. EXTRA: Visualizing Spatial Measurements with Show Plots



The **Show plots** function in the **measurement table** can be used to explore the **spatial distribution** of the classified **cell populations**. In this exercise, we will visualize:

1. The signed **distance** of **TCells** and **CD8+TCells** to the **tumor border**.
2. The nearest-neighbor **distances** between the **two target cell populations**.

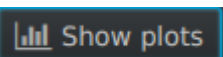
The **populations** of interest are:

- **TCells:** CD45+:CD3+:CD8–
- **CD8+TCells:** CD45+:CD3+:CD8+

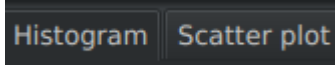
Important: Complete the **Signed distance to annotation 2D** and **Detection centroid distances 2D** analyses before starting this exercise. Otherwise, the required distance measurements will not be available in the detection measurement table.

Visualize the Distance to the Tumor Border

- Go to **Measure** > **Show detection measurements**.
- Confirm that the **table** contains the column '**Signed distance to annotation Tumor_ML µm**'.



- Click **Show plots** at the top of the measurement table.



- Select the **Histogram** tab
- From the measurement menu, select: '**Signed distance to annotation Tumor_ML µm**'

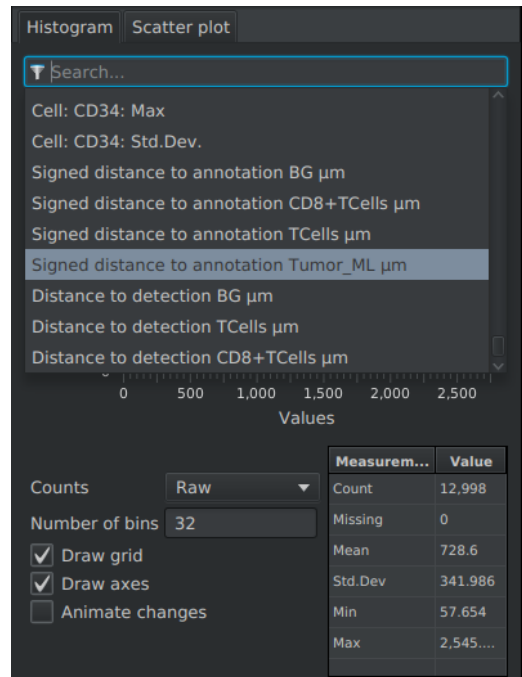


Figure 61. Selection of measurement 'Signed distance to annotation Tumor_ML μm ' for histogram.

- The **histogram** will display the distribution of **cell distances** relative to the **boundary** of the **Tumor_ML** annotation:
 - **Negative values:** cell centroids located inside the tumor.
 - **Values close to zero:** cell centroids located at or near the tumor border.
 - **Positive values:** cell centroids located outside the tumor (surrounding microenvironment).

Note: The **vertical axis** displays the **number of cells** within each distance interval, while the **horizontal axis** displays the signed **distance to the tumor border** in micrometres.

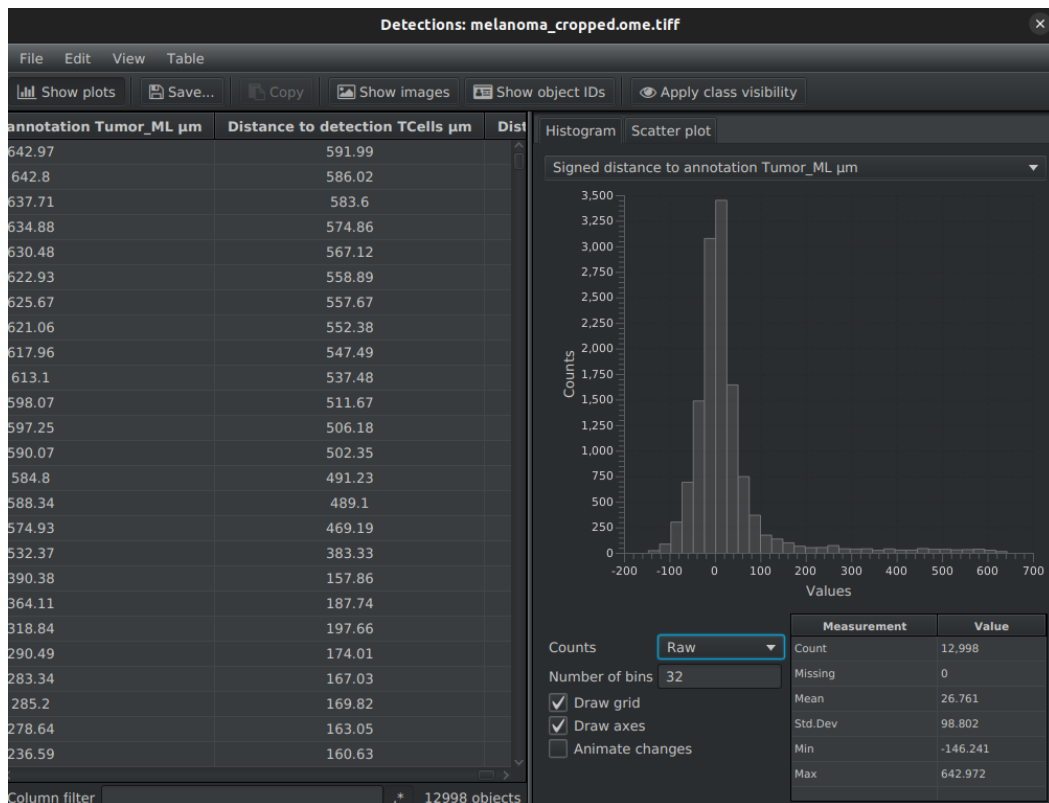


Figure 62. Detection table and histogram plot showing signed cell distances relative to the Tumor_ML boundary.

- Change the **Number of bins** setting.
- Compare a lower number of bins, such as **15–20**, with a higher number, such as **40–50**. Observe how the representation of the distance distribution changes.

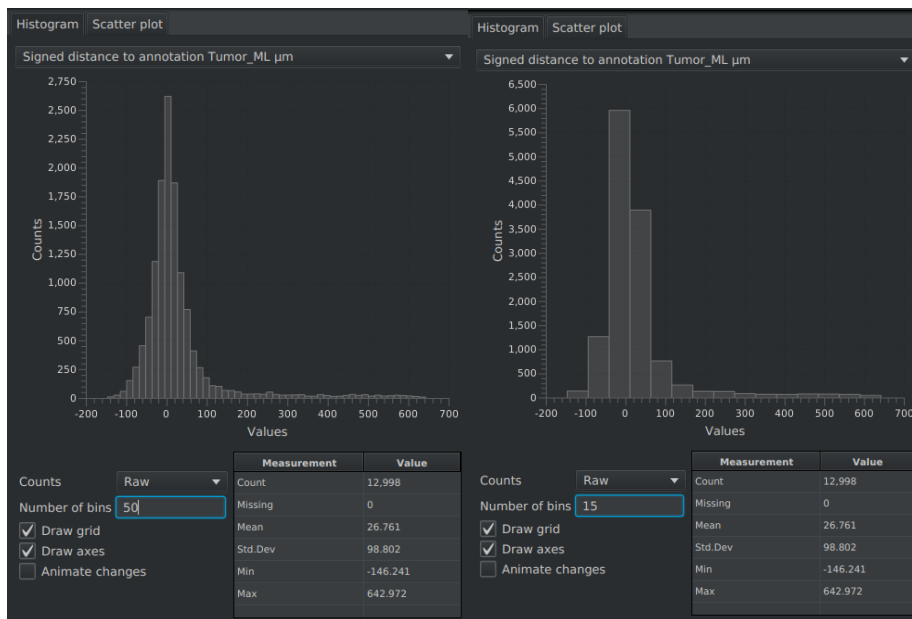


Figure 63. Histogram of signed distances from classified cells to the Tumor_ML border.

Note: Fewer bins provide a simplified overview of the distribution. More bins reveal finer details but can make the histogram appear more irregular. Changing the number of bins does not change the measurements, only how the distribution is displayed.

Compare TCells and CD8+ TCells

- Go to **Class list** panel and locate **TCells** and **CD8+TCells** and **BG** classes to be analyzed.
- Hide the rest of the classes and left visible just **TCells** population.

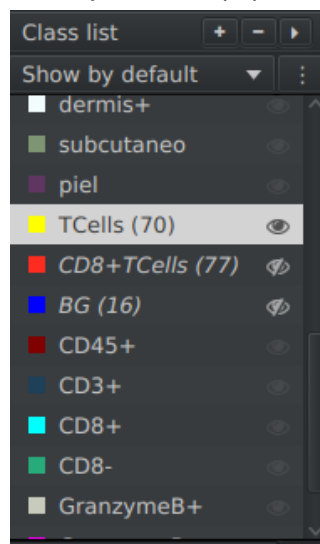


Figure 64. Hide the rest of the classes and left visible just TCells population.

- Use **Apply class visibility**  **Apply class visibility** to show only **TCells** on the measurement table.
- Inspect the histogram of distances to the **Tumor_ML** border, by selecting: '**Signed distance to annotation Tumor_ML µm**'

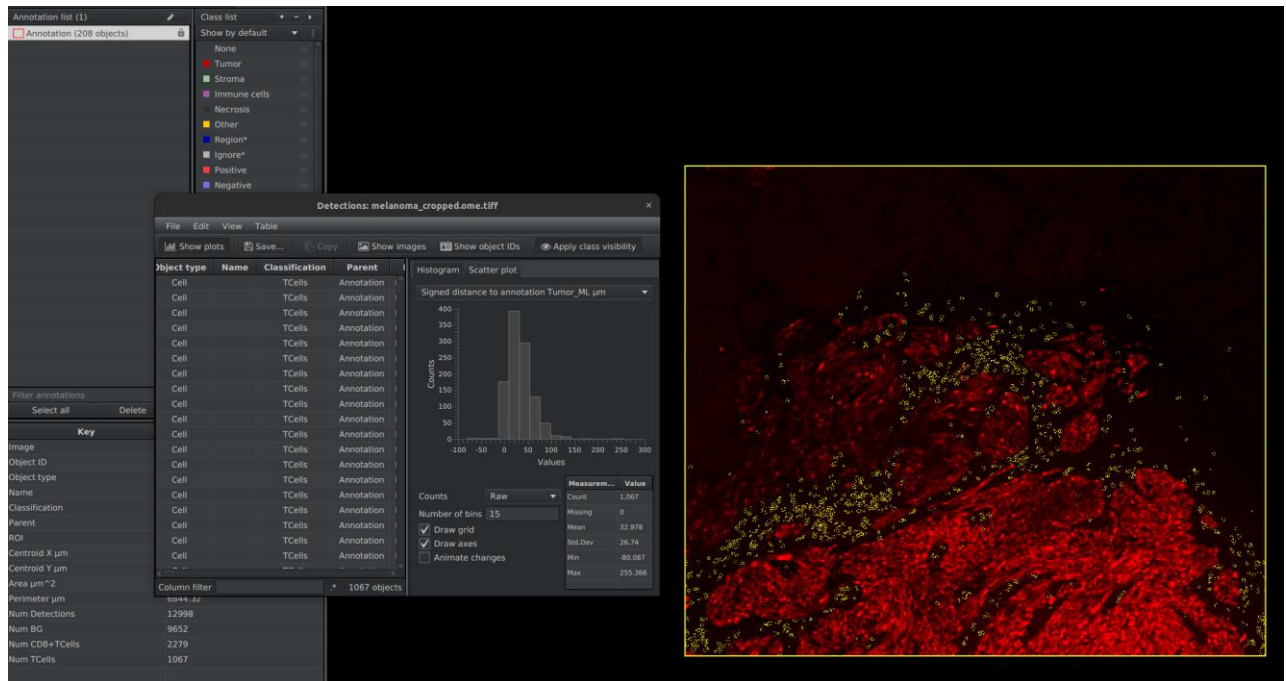


Figure 65. Histogram plot showing the TCells distribution for Signed distance to annotation Tumor_ML.

- Repeat the process for the **CD8+TCells** population.

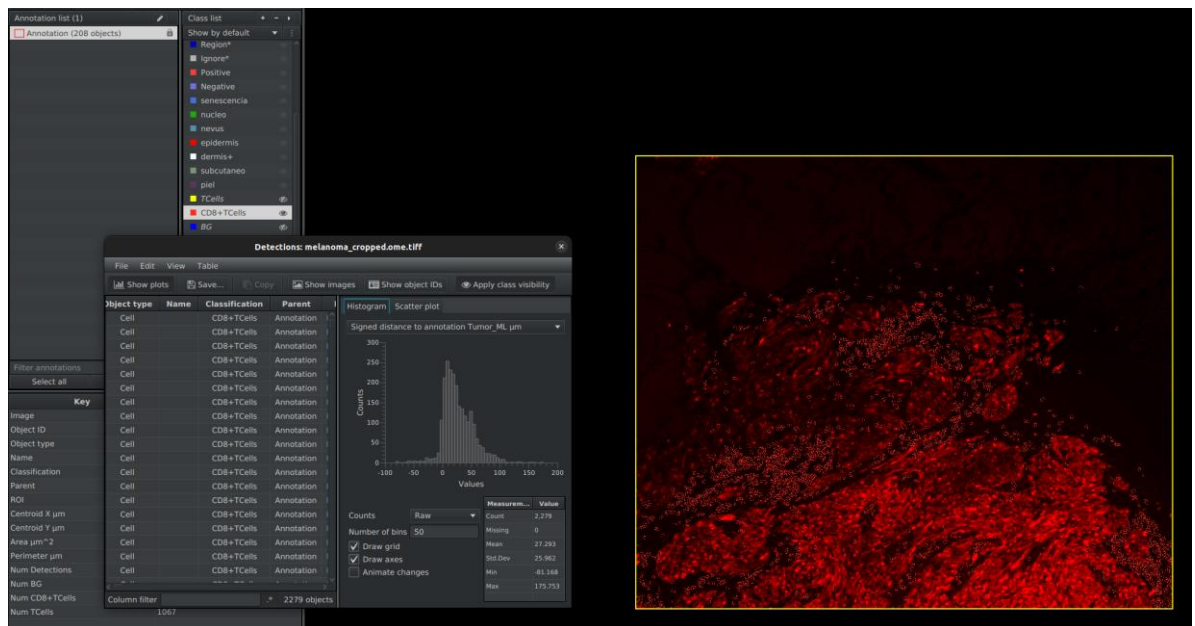


Figure 66. Histogram plot showing the CD8+TCells distribution for Signed distance to annotation Tumor_ML.

- Compare the two distance distributions.

Important: A distribution shifted towards negative values suggests greater localization inside the tumor compartment. A distribution concentrated around zero suggests accumulation near the tumor boundary. A distribution shifted towards positive values suggests that more cells are located outside the tumor.

9. EXTRA: Visualize Distances between Cell populations

Explore TCells to CD8+ TCells

- Restrict the displayed detections to **TCells** population.

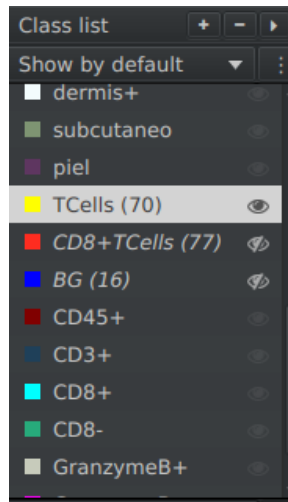


Figure 67. Hide the rest of the classes and leave visible just TCells population.

- In the **histogram**, select the measurement '**Distance to detection CD8+ TCells μm** '

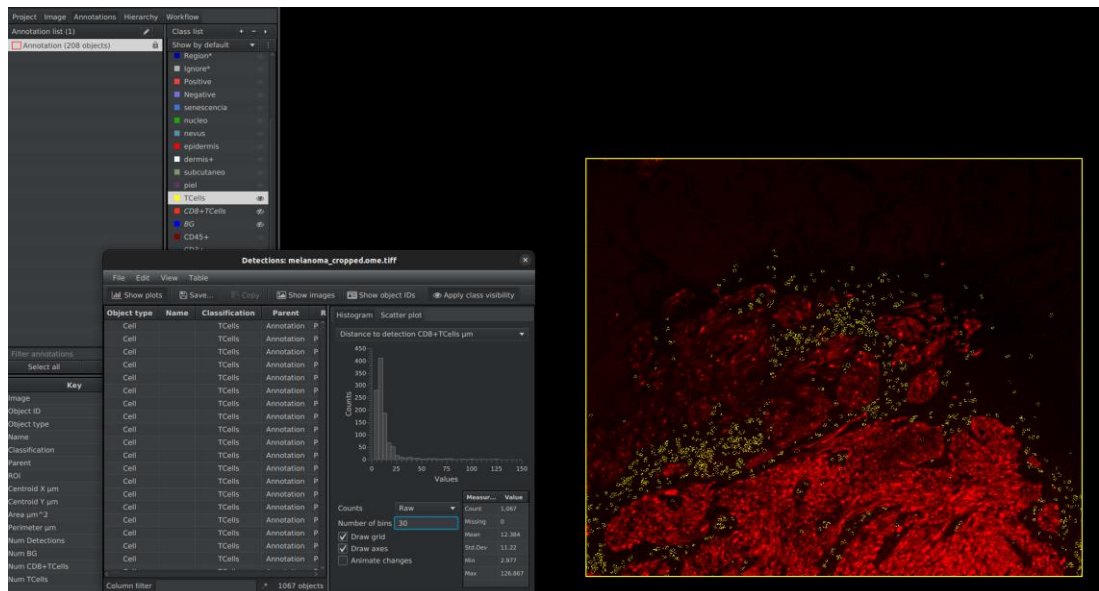


Figure 68. Histogram plot showing the TCells distribution for distance to CD8+TCells.

- Inspect the resulting distributions: **Shorter** distance values indicate that a **TCell** has a **nearby CD8+TCell**, whereas **larger** values indicate **greater** spatial separation.
- Repeat the process for the **CD8+TCells** population.

10. EXTRA: Create a Scatter Plot

The **Scatter plot** option can be used to examine the **relationship** between the **position** of a **cell** relative to the **tumor** and its **proximity** to another cell **population**.

- Open **Show plots**.
- Select the **Scatter plot** tab.
- Restrict the displayed detections to **TCells** population:
 - X-axis:** '**Signed distance to annotation Tumor_ML μm** '
 - Y-axis:** '**Distance to detection CD8+TCells μm** '

- Each point represents one **TCell**. The scatter plot combines the cell **position** relative to the tumor border with the proximity to the nearest **CD8+TCell**:
 - X-axis:** Signed distance from the **TCell** centroid to the **Tumor_ML** border.
 - Negative values indicate **TCells** located inside the tumor.
 - Values close to zero indicate **TCells** located near the tumor border.
 - Positive values indicate **TCells** located outside the tumor.
 - Y-axis:** Distance from each **TCell** to the nearest **CD8+TCell**.
 - Lower values indicate greater proximity between the two populations.
 - Higher values indicate greater spatial separation.
- In this example, most **TCells** are located outside the tumor and relatively close to the nearest **CD8+TCell**. A small number of cells show larger distances and may represent spatially isolated cells or potential segmentation or classification outliers.

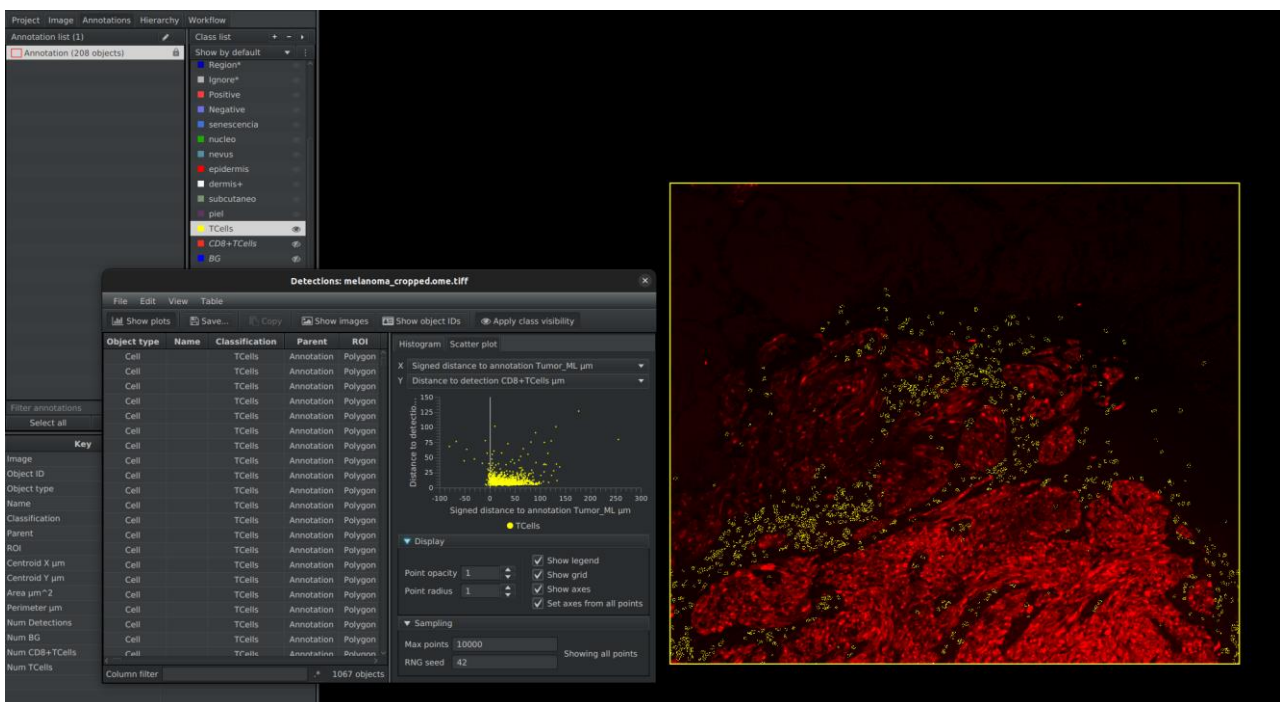


Figure 69. Scatter plot relating T-cell position around the tumor border to proximity to the nearest CD8+TCell.